Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #211 *Topic 1*

A company hosts multiple production applications. One of the applications consists of resources from Amazon EC2, AWS Lambda, Amazon RDS, Amazon Simple Notification Service (Amazon SNS), and Amazon Simple Queue Service (Amazon SQS) across multiple AWS Regions. All company resources are tagged with a tag name of “application” and a value that corresponds to each application. A solutions architect must provide the quickest solution for identifying all of the tagged components.

Which solution meets these requirements?

A. Use AWS CloudTrail to generate a list of resources with the application tag.

B. Use the AWS CLI to query each service across all Regions to report the tagged components.

C. Run a query in Amazon CloudWatch Logs Insights to report on the components with the application tag.

D. Run a query with the AWS Resource Groups Tag Editor to report on the resources globally with the application tag. **Most Voted**

Correct Answer: *D*

Community vote distribution

D (100%)

Comments

cookieMr Highly Voted 1 year, 11 months ago

Selected Answer: D

A is not the quickest solution because CloudTrail primarily focuses on capturing and logging API activity. While it can provide information about resource changes, it may not provide a comprehensive and quick way to identify all the tagged components across multiple services and Regions.

B involves manually querying each service using the AWS CLI, which can be time-consuming and cumbersome, especially when dealing with multiple services and Regions. It is not the most efficient solution for quickly identifying tagged components.

C is focused on analyzing logs rather than directly identifying the tagged components. While CloudWatch Logs Insights can help extract information from logs, it may not provide a straightforward and quick way to gather a consolidated list of all tagged components across different services and Regions.

D is the quickest solution as it leverages the Resource Groups Tag Editor, which is specifically designed for managing and organizing resources based on tags. It offers a centralized and efficient approach to generate a report of tagged components across multiple services and Regions.

upvoted 25 times

upvoted 25 times

satyaammm Most Recent 3 months, 3 weeks ago

Selected Answer: D

D is the most suitable here.

upvoted 1 times

JA2018 6 months, 3 weeks ago

Selected Answer: D

To use the AWS Resource Groups Tag Editor to locate resources with specific tags, you can:

1. Open the Tag Editor console
2. Select the AWS regions to search in
3. Choose a resource type
4. Enter a tag key or key and value pair in the Tags fields
5. Select Add or press Enter to finish

https://docs.aws.amazon.com/tag-editor/latest/userguide/find-resources-to-tag.html#:~:text=To%20find%20resources%20to%20tag,are%20returned%20by%20the%20query.

upvoted 2 times

TariqKipkemei 1 year, 8 months ago

Selected Answer: D

Tags are key and value pairs that act as metadata for organizing your AWS resources

upvoted 3 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: D

D. Run a query with the AWS Resource Groups Tag Editor to report on the resources globally with the application tag

upvoted 4 times

Bmarodi 2 years ago

Selected Answer: D

A solutions architect can provide the quickest solution for identifying all of the tagged components by running running a query with the AWS Resource Groups Tag Editor to report on the resources globally with the application tag, hence the option D is right answer.

upvoted 3 times

Dondozzy 2 years, 2 months ago

Selected Answer: D

The answer is D

upvoted 3 times

sh0811 2 years, 4 months ago

Selected Answer: D

D□ □□□□.

upvoted 2 times

Training4aBetterLife 2 years, 4 months ago

Selected Answer: D

https://docs.aws.amazon.com/tag-editor/latest/userguide/tagging.html

upvoted 4 times

Rudraman 2 years, 4 months ago

Answer is D.

upvoted 2 times

techhb 2 years, 4 months ago

Selected Answer: D

validated
https://docs.aws.amazon.com/tag-editor/latest/userguide/tagging.html
upvoted 2 times

kbaruu 2 years, 4 months ago

Selected Answer: D

D is correct
upvoted 2 times

waiyiu9981 2 years, 4 months ago

Selected Answer: D

https://www.examtopics.com/discussions/amazon/view/51352-exam-aws-certified-solutions-architect-associate-saa-c02/
upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #212 *Topic 1*

A company needs to export its database once a day to Amazon S3 for other teams to access. The exported object size varies between 2 GB and 5 GB. The S3 access pattern for the data is variable and changes rapidly. The data must be immediately available and must remain accessible for up to 3 months. The company needs the most cost-effective solution that will not increase retrieval time.

Which S3 storage class should the company use to meet these requirements?

A. S3 Intelligent-Tiering **Most Voted**

B. S3 Glacier Instant Retrieval

C. S3 Standard

D. S3 Standard-Infrequent Access (S3 Standard-IA)

Correct Answer: *A*

Community vote distribution

A (77%) D (10%) Other (13%)

Comments

techhb Highly Voted 2 years, 4 months ago

Selected Answer: A

S3 Intelligent-Tiering monitors access patterns and moves objects that have not been accessed for 30 consecutive days to the Infrequent Access tier and after 90 days of no access to the Archive Instant Access tier.

upvoted 21 times

Devsin2000 2 years, 1 month ago

https://aws.amazon.com/getting-started/hands-on/getting-started-using-amazon-s3-intelligent-tiering/

upvoted 6 times

pentium75 Highly Voted 1 year, 5 months ago

Selected Answer: A

I think it cannot be clearly answered because we know that the 'access pattern is variable and changes rapidly', but ultimately it depends on the total number and volume of accesses. All four options meet the "not increase retrieval time" requirement (even Glacier Instant Retrieval has "the same latency and access time as S3 Standard"). If data would be rarely accessed, B would be cheapest. If it would be constantly accessed, C would be cheapest (we'd pay the Intelligent Tiering fee but it would never move anything to a cheaper tier). Inbetween it would be D.

But I guess the key is Amazon's clear recommendation to use Intelligent Tiering (A) for "unknown or changing access" patterns, which matches the statement in the question.

upvoted 10 times

satyaammm Most Recent 3 months, 3 weeks ago

Selected Answer: A

S3 intelligent tiering is the most suitable for unpredictable workloads.

upvoted 1 times

iamroyalty_k 4 months, 1 week ago

Selected Answer: A

S3 Intelligent-Tiering dynamically optimizes storage costs for changing access patterns, ensuring immediate availability and cost-efficiency, which makes it the best fit for the given requirements.

Why not the other options?

B. S3 Glacier Instant Retrieval:
While it provides low retrieval latency, Glacier storage classes are optimized for archival data with infrequent access, not for variable and potentially high-frequency access patterns.

C. S3 Standard:
This provides immediate availability but is more expensive than Intelligent-Tiering for data with unpredictable access patterns.

D. S3 Standard-Infrequent Access (S3 Standard-IA):
S3 Standard-IA is optimized for data that is accessed infrequently but requires immediate availability. However, if the access pattern is variable or frequent, it could lead to higher costs due to retrieval and access fees.

upvoted 1 times

skybrink 4 months, 2 weeks ago

Selected Answer: D

A. S3 Intelligent-Tiering:

Intelligent-Tiering automatically moves data between storage tiers based on access patterns. However, it incurs additional monitoring and automation costs. For data with a short, predictable retention period (3 months), S3 Standard-IA is more cost-effective.
B. S3 Glacier Instant Retrieval:

Although Glacier Instant Retrieval offers low-latency access at a lower cost than Standard-IA, it is more suited for archival workloads where retrieval is rare. It is not ideal for workloads with variable access patterns like this one.
C. S3 Standard:

S3 Standard provides low-latency access, but it is more expensive than S3 Standard-IA for infrequently accessed data. It is better suited for frequently accessed data.

upvoted 2 times

ensbrvsnss 9 months, 2 weeks ago

Selected Answer: B

Instant

upvoted 1 times

lofzee 1 year ago

Selected Answer: A

unknown / changing access patterns = intelligent tiering. memorise

upvoted 4 times

ManikRoy 1 year, 1 month ago

Selected Answer: A

Unpredictable access pattern - Intelligent tiering

upvoted 3 times

MehulKapadia 1 year, 1 month ago

Selected Answer: A

Correct Answer A:
When data access pattern is not known then Intelligent-tiering can help by monitoring data-access pattern and move object internally accordingly an still ensure faster retrieval. Also There is no object retrieval fees/changes for S3 Intelligent Tier(So cost savings).

Option C is not a valid answer because name itself says Infrequent Access(IA): S3 Standard-IA is for data that is accessed less frequently.

upvoted 3 times

Uzbekistan 1 year, 2 months ago

Selected Answer: C

Immediate Availability: S3 Standard provides immediate access to the data upon upload. This ensures that the exported database is immediately available for other teams to access without any retrieval delays.

Variable Access Pattern: S3 Standard is designed to handle variable access patterns efficiently. It can accommodate rapid changes in access patterns without any impact on performance or latency.

Retention Period: S3 Standard is suitable for storing data that needs to remain accessible for up to 3 months. It does not have any retrieval fees or delays, making it ideal for this scenario where immediate access is required.

upvoted 3 times

escalibran 1 year, 2 months ago

Feels like half the scenario or answers are missing. Where's the "remove objects after 90 days"? Intelligent Tiering has an upcharge for the provided convenience - does it even make sense, when objects won't remain long enough to be archived? Other classes trade storage cost for request costs. Dependent on how often objects are queried, IA might make sense. Even Glacier Instant Retrieval could come out ahead, given minimal access (and it has 90 days minimum storage duration, exact fit for the description).

With no further details provided, this is just throwing darts blindly.

upvoted 1 times

escalibran 1 year, 2 months ago

Given just the uncertain access patterns AND limited storage time, I would argue in favor of simple S3 Standard.
If the question mentioned that the pattern of access varies across objects, but is relatively consistent for the individual objects, intelligent tiering may be worth it. Otherwise you just pay more to have objects monitored for Infrequent Access, and then suddenly become popular after being moved.

upvoted 2 times

MrPCarrot 1 year, 3 months ago

A is the perfect answer - The S3 access pattern for the data is variable and changes rapidly.

upvoted 2 times

bujuman 1 year, 4 months ago

Selected Answer: A

With regard to "The S3 access pattern for the data is variable and changes rapidly"
Even though Answer B cooudl fifull some requirements, Answer A is For long-lived data that have unpredictable access patterns.

upvoted 2 times

theochan 1 year, 4 months ago

Selected Answer: B

"immediately available" =>
D is not immediately, and for cost B < A/C

upvoted 3 times

VladanO 1 year, 7 months ago

Selected Answer: B
https://aws.amazon.com/s3/storage-classes/glacier/instant-retrieval/
"Amazon S3 Glacier Instant Retrieval is an archive storage class that delivers the lowest-cost storage for long-lived data that is rarely accessed and requires retrieval in milliseconds"

upvoted 3 times

ivan_riqueros12 1 year, 7 months ago

Selected Answer: A

A. El patrón de acceso a los datos es variable y cambia rápidamente = S3 Intelligent-Tiering

upvoted 1 times

Abdou1604 1 year, 7 months ago

very important note , S3 Intelligent-Tiering got no retrival charges

upvoted 4 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #213 *Topic 1*

A company is developing a new mobile app. The company must implement proper traffic filtering to protect its Application Load Balancer (ALB) against common application-level attacks, such as cross-site scripting or SQL injection. The company has minimal infrastructure and operational staff. The company needs to reduce its share of the responsibility in managing, updating, and securing servers for its AWS environment.

What should a solutions architect recommend to meet these requirements?

A. Configure AWS WAF rules and associate them with the ALB. **Most Voted**

B. Deploy the application using Amazon S3 with public hosting enabled.

C. Deploy AWS Shield Advanced and add the ALB as a protected resource.

D. Create a new ALB that directs traffic to an Amazon EC2 instance running a third-party firewall, which then passes the traffic to the current ALB.

Correct Answer: *A*

Community vote distribution

A (79%) C (21%)

Comments

cookieMr Highly Voted 1 year, 11 months ago

Selected Answer: A

By configuring AWS WAF rules and associating them with the ALB, the company can filter and block malicious traffic before it reaches the application. AWS WAF offers pre-configured rule sets and allows custom rule creation to protect against common vulnerabilities like XSS and SQL injection.

Option B does not provide the necessary security and traffic filtering capabilities to protect against application-level attacks. It is more suitable for hosting static content rather than implementing security measures.

Option C is focused on DDoS protection rather than application-level attacks like XSS or SQL injection. While AWS Shield Advanced does not address the specific requirements mentioned in the scenario.

Option D involves maintaining and securing additional infrastructure, which goes against the requirement of reducing responsibility and relying on minimal operational staff.

upvoted 19 times

ShinobiGrappler Highly Voted 2 years, 4 months ago

Selected Answer: C

C --- Read and understand the question. *The company needs to reduce its share of responsibility in managing, updating, and securing servers for its AWS environment* Go with AWS Shield advanced --This is a managed service that includes AWS WAF, custom mitigations, and DDoS insight.

upvoted 19 times

Guru4Cloud 1 year, 9 months ago

I dont know how this comment gets 11x upvotes.
A.To filter traffic and protect against application attacks like cross-site scripting and SQL injection, the company can use AWS Web Application Firewall with managed rules on the Application Load Balancer. This provides security with minimal infrastructure and operations overhead.

upvoted 30 times

pentium75 1 year, 5 months ago

Read the understand the question. The company needs protection "against common application-level (!) attacks" which is provided by a Layer 7 service like WAF. AWS Shield Advanced protects from network-level (!) attacks.

upvoted 8 times

AWSSURI 9 months, 2 weeks ago

Have you read and understood the question first!! It says application-level attacks such as cross -site scripting, SQL injection which automatically points to AWS WAF
Go to this link and look at AWS WAF benefits https://docs.aws.amazon.com/waf/latest/developerguide/what-is-aws-waf.html

upvoted 4 times

abriggy 10 months, 2 weeks ago

WRONG. Answer is A. Don't let all these upvotes fool you

upvoted 4 times

satyaammm Most Recent 3 months, 3 weeks ago

Selected Answer: A

AWS WAF is the most suitable for Application Level attacks like Cross Site Scripting and SQL Injection attacks.

upvoted 1 times

toyaji 9 months, 3 weeks ago

Selected Answer: C

Most of all, A and C are both available technically, right? So the point of question is not about technical posibility. Its about "share of the respoinsibility" which is intended to ask of which service provides "Support Plan" - AWS Shield Response Team (SRT)

upvoted 2 times

ChymKuBoy 11 months, 2 weeks ago

Selected Answer: A

A for sure

upvoted 2 times

a7md0 11 months, 3 weeks ago

Selected Answer: A

AWS Shield Advanced for DDoS Attacks and not SQL injection which is protected by AWS WAF

upvoted 2 times

ManikRoy 1 year, 1 month ago

Selected Answer: A

AWS WAF with managed rules.

upvoted 2 times

Solomon2001 1 year, 1 month ago

Explanation:

Option A:
AWS WAF (Web Application Firewall) provides protection against common web exploits by allowing you to create rules that block common attack patterns such as SQL injection and cross-site scripting (XSS).
By associating AWS WAF rules with the ALB, you can protect your application from these types of attacks without managing, updating, and securing servers yourself.
AWS WAF is a managed service, so it reduces the operational overhead for the company.

Option C:

AWS Shield Advanced provides DDoS protection, but it doesn't include application-level protection like AWS WAF does.
upvoted 3 times

sandordini 1 year, 1 month ago

If you read SQL Injection, Cross-site scripting >>> Always look for: WAF
upvoted 2 times

bujuman 1 year, 4 months ago

Selected Answer: A

This is confusing "The company needs to reduce its share of the responsibility in managing, updating, and securing servers for its AWS environment." But could be acheived when using WAF and AWS managed Rules.
upvoted 3 times

thewalker 1 year, 4 months ago

Selected Answer: A

A is the answer.
upvoted 2 times

farnamjam 1 year, 4 months ago

Selected Answer: A

AWS Shield Advanced does not directly protect against XSS (cross-site scripting) or SQL injection attacks. It focuses on defending against Distributed Denial of Service (DDoS) attacks, which aim to overwhelm resources and disrupt availability.
upvoted 2 times

awsgeek75 1 year, 5 months ago

Selected Answer: A

S makes more sense as Shield Advanced (which actually contains WAF) doesn't provide any additional benefits apart from networks protection. WAF will still have to be configured. So just use WAF to fulfil the requirements.
upvoted 3 times

pentium75 1 year, 5 months ago

Selected Answer: A

You need to "configure AWS WAF rules and associate them with the ALB" which is A. AWS Shield Advance INTEGRATES with WAF, so you can manage WAF through Shield Advanced, but still you would need to set it up and configure rules, which C does not mention.
upvoted 5 times

[Removed] 1 year, 5 months ago

AWS Shield is not only DDos and it handle Layer 3 and layer 4 including AWS WAF so C should match.
upvoted 1 times

pentium75 1 year, 5 months ago

"Shield Advanced provides ... integration (!) with AWS WAF", but you still need WAF. And you need WAF rules, whereever you configure them.
upvoted 2 times

TariqKipkemei 1 year, 8 months ago

Selected Answer: A

AWS WAF helps you protect against common web exploits and bots that can affect availability, compromise security, or
consume excessive resources. Protect against vulnerabilities and exploits such as SQL injection or Cross site scripting attacks

consume excessive resources. Protect against vulnerabilities and exploits such as SQL injection or Cross site scripting attacks.

upvoted 6 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: A

To filter traffic and protect against application attacks like cross-site scripting and SQL injection, the company can use AWS Web Application Firewall with managed rules on the Application Load Balancer. This provides security with minimal infrastructure and operations overhead.

upvoted 4 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #214 *Topic 1*

A company's reporting system delivers hundreds of .csv files to an Amazon S3 bucket each day. The company must convert these files to Apache Parquet format and must store the files in a transformed data bucket.

Which solution will meet these requirements with the LEAST development effort?

A. Create an Amazon EMR cluster with Apache Spark installed. Write a Spark application to transform the data. Use EMR File System (EMRFS) to write files to the transformed data bucket.

B. Create an AWS Glue crawler to discover the data. Create an AWS Glue extract, transform, and load (ETL) job to transform the data. Specify the transformed data bucket in the output step. Most Voted

C. Use AWS Batch to create a job definition with Bash syntax to transform the data and output the data to the transformed data bucket. Use the job definition to submit a job. Specify an array job as the job type.

D. Create an AWS Lambda function to transform the data and output the data to the transformed data bucket. Configure an event notification for the S3 bucket. Specify the Lambda function as the destination for the event notification.

Correct Answer: *B*

Community vote distribution

B (100%)

Comments

Babba Highly Voted 1 year, 10 months ago

Selected Answer: B

It looks like AWS Glue allows fully managed CSV to Parquet conversion jobs: https://docs.aws.amazon.com/prescriptive-guidance/latest/patterns/three-aws-glue-etl-job-types-for-converting-data-to-apache-parquet.html

upvoted 20 times

awsgeek75 11 months, 1 week ago

A text book use case: https://docs.aws.amazon.com/prescriptive-guidance/latest/patterns/three-aws-glue-etl-job-types-for-converting-data-to-apache-parquet.html#three-aws-glue-etl-job-types-for-converting-data-to-apache-parquet-epics

B is the correct answer.

upvoted 2 times

cookieMr Highly Voted 1 year, 5 months ago

Selected Answer: B

AWS Glue is a fully managed ETL service that simplifies the process of preparing and transforming data for analytics. Using AWS Glue requires minimal development effort compared to the other options.

Option A requires more development effort as it involves writing a Spark application to transform the data. It also introduces additional infrastructure management with the EMR cluster.

Option C requires writing and managing custom Bash scripts for data transformation. It requires more manual effort and does not provide the built-in capabilities of AWS Glue for data transformation.

Option D requires developing and managing a custom Lambda for data transformation. While Lambda can handle the transformation, it requires more effort compared to AWS Glue, which is specifically designed for ETL operations.

Therefore, option B provides the easiest and least development effort by leveraging AWS Glue's capabilities for data discovery, transformation, and output to the transformed data bucket.

upvoted 10 times

satyaammm Most Recent 3 months, 3 weeks ago

Selected Answer: B

AWS Glue is designed for ETL and this scenario.

upvoted 1 times

iamroyalty_k 4 months, 1 week ago

Selected Answer: B

AWS Glue offers a serverless, automated, and cost-effective solution with minimal development and operational effort, making it the best choice for this use case.

Why not the other options?
A. Amazon EMR cluster with Apache Spark:
While EMR and Spark can handle this task, it requires more setup, maintenance, and development effort compared to AWS Glue. Managing the cluster introduces operational overhead.

C. AWS Batch with Bash job definition:
Using AWS Batch for this would require creating custom Bash scripts for the transformation and managing jobs. This introduces more complexity and development effort than AWS Glue

D. AWS Lambda with S3 event notifications:
Lambda is suitable for lightweight, real-time processing. However, converting hundreds of .csv files into Parquet format could exceed Lambda's execution time and resource limits, leading to scalability challenges.

upvoted 1 times

lofzee 6 months, 1 week ago

Selected Answer: B

AWS Glue and parquet go hand in hand

upvoted 2 times

zinabu 8 months, 1 week ago

i will go with answer B cause: You can use AWS Glue to write ETL jobs in a Python shell environment. You can also create both batch and streaming ETL jobs by using Python (PySpark) or Scala in a managed Apache Spark environment.
Apache Parquet is built to support efficient compression and encoding schemes. It can speed up your analytics workloads because it stores data in a columnar fashion. Converting data to Parquet can save you storage space, cost, and time in the longer run

upvoted 3 times

Rido4good 10 months, 3 weeks ago

D
I think people are forgetting the question says "Low Overhead".

upvoted 1 times

awsgeek75 10 months, 3 weeks ago

Pray tell, how is a Lambda less overhead than B or even A?

upvoted 3 times

nileeka97 1 year, 2 months ago

Selected Answer: B

Parquet format ========> Amazon Glue

upvoted 4 times

Guru4Cloud 1 year, 2 months ago

Selected Answer: B

B. Create an AWS Glue crawler to discover the data. Create an AWS Glue extract, transform, and load (ETL) job to transform the data. Specify the transformed data bucket in the output step.

upvoted 3 times

markw92 1 year, 5 months ago

Least development effort means lambda. Glue also works but more overhead and cost. A simple lambda like this
https://github.com/ayshaysha/aws-csv-to-parquet-converter/blob/main/csv-parquet-converter.py
can be used to convert as soon as you see files in s3 bucket.

upvoted 4 times

achevez85 1 year, 9 months ago

Selected Answer: B

https://docs.aws.amazon.com/prescriptive-guidance/latest/patterns/three-aws-glue-etl-job-types-for-converting-data-to-apache-parquet.html

upvoted 3 times

Training4aBetterLife 1 year, 10 months ago

Selected Answer: B

S3 provides a single control to automatically encrypt all new objects in a bucket with SSE-S3 or SSE-KMS. Unfortunately, these controls only affect new objects. If your bucket already contains millions of unencrypted objects, then turning on automatic encryption does not make your bucket secure as the unencrypted objects remain.

For S3 buckets with a large number of objects (millions to billions), use Amazon S3 Inventory to get a list of the unencrypted objects, and Amazon S3 Batch Operations to encrypt the large number of old, unencrypted files.

upvoted 2 times

Training4aBetterLife 1 year, 10 months ago

Versioning:

When you overwrite an S3 object, it results in a new object version in the bucket. However, this will not remove the old unencrypted versions of the object. If you do not delete the old version of your newly encrypted objects, you will be charged for the storage of both versions of the objects.

S3 Lifecycle

If you want to remove these unencrypted versions, use S3 Lifecycle to expire previous versions of objects. When you add a Lifecycle configuration to a bucket, the configuration rules apply to both existing objects and objects added later. C is missing this step, which I believe is what makes B the better choice. B includes the functionality of encrypting the old unencrypted objects via Batch Operations, whereas, Versioning does not address the old unencrypted objects.

upvoted 1 times

Training4aBetterLife 1 year, 10 months ago

Please delete this. I was meaning to place this response on a different question.

upvoted 2 times

Training4aBetterLife 1 year, 10 months ago

Please delete this. I was meaning to place this response on a different question.

upvoted 1 times

Rudraman 1 year, 10 months ago

ETL = Glue

upvoted 4 times

Aninina 1 year, 10 months ago

Selected Answer: B

B is the correct answer

upvoted 2 times

techhb 1 year, 10 months ago

Selected Answer: B

AWS Glue Crawler is for ETL

upvoted 2 times

kbaruu 1 year, 10 months ago

Selected Answer: B

The correct answer is B

upvoted 2 times

Mamiololo 1 year, 10 months ago

B is the answer

upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #215 *Topic 1*

A company has 700 TB of backup data stored in network attached storage (NAS) in its data center. This backup data need to be accessible for infrequent regulatory requests and must be retained 7 years. The company has decided to migrate this backup data from its data center to AWS. The migration must be complete within 1 month. The company has 500 Mbps of dedicated bandwidth on its public internet connection available for data transfer.

What should a solutions architect do to migrate and store the data at the LOWEST cost?

A. Order AWS Snowball devices to transfer the data. Use a lifecycle policy to transition the files to Amazon S3 Glacier Deep Archive. **Most Voted**

B. Deploy a VPN connection between the data center and Amazon VPC. Use the AWS CLI to copy the data from on premises to Amazon S3 Glacier.

C. Provision a 500 Mbps AWS Direct Connect connection and transfer the data to Amazon S3. Use a lifecycle policy to transition the files to Amazon S3 Glacier Deep Archive.

D. Use AWS DataSync to transfer the data and deploy a DataSync agent on premises. Use the DataSync task to copy files from the on-premises NAS storage to Amazon S3 Glacier.

Correct Answer: *A*

Community vote distribution

A (98%) B

Comments

voccer Highly Voted 1 year, 11 months ago

Selected Answer: A

hundreds of Terabytes => always use Snowball

upvoted 14 times

TariqKipkemei Highly Voted 1 year, 8 months ago

Selected Answer: A

Terabytes, low costs, limited time = AWS snowball devices

upvoted 10 times

sjb009 Most Recent 2 months, 3 weeks ago

Selected Answer: A

70TB = AWS Snowball

upvoted 1 times

Rcosmos 4 months, 3 weeks ago

Selected Answer: B

Resumo:
A Opção B é a solução mais eficiente, pois:

Garante a criptografia automática de todos os objetos futuros com configurações de criptografia padrão no bucket.
Usa o Inventário do S3 e trabalhos de operações em lote para criptografar objetos existentes sem esforço manual significativo ou movimentação de dados desnecessária.

upvoted 1 times

awsgeek75 1 year, 5 months ago

It took me quite some time to do the mental math for realising that the data can't be transferred in 30 days. Also, note the MBps (Megabits) and not Megabytes. 500Mbps is like 60MBps. That's a lame connection to transfer anything!

upvoted 7 times

lofzee 1 year ago

well.. standard internet connections are measured in Mbps and 500 Mbps is pretty decent to be fair (by english standards).
even still at that speed it would take you about 4 months to upload 700 TB. So the only option here is to use snowball.
Answer is A

upvoted 3 times

pentium75 1 year, 5 months ago

Selected Answer: A

B and D would use existing 500 mbps Internet connection which cannot transfer more than ca. 160 TB in a month. C would cost a lot, take weeks to deliver, and still not provide more bandwidth. Thus A is the simply the only option, thus also the one with "lowest cost".

upvoted 3 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: A

A. Order AWS Snowball devices to transfer the data. Use a lifecycle policy to transition the files to Amazon S3 Glacier Deep Archive.

upvoted 2 times

gosai90786 1 year, 11 months ago

one DataSync agent can use 10GBps and can setup a bandwidth.
So total time = (700X1000)GB/10GBps = 70000 sec = 19.4 days.
Using Multiple Snowball devices will involve ordering them from AWS, setting them up on your data-center for copy and then incurring the shipping cost for too and fro movement to your AWS cloud.
if time constraint was critical , say 1 week then snowball would have been a viable option. But here we have 30 days, so DataSync will be less costly(takes `19days)

upvoted 2 times

JA2018 6 months, 3 weeks ago

From Google search: "do I need to pay to ship back aws snow devices back to AWS?"

No, you don't need to pay to ship back an AWS Snowball Edge device to AWS. The device comes with a prepaid UPS shipping label on the E Ink display that contains the correct address for return. You can arrange for UPS to pick up the device or drop it off at a UPS package drop-off facility.

upvoted 1 times

slackbot 1 year, 9 months ago

your math is wrong mate, and they have 0.5Gbps connection, not 10GBps
500Mpbs = roughly 60MBps
30x24x3600x0.06TB = roughly 155TB
this is way short of 700TB

this is way short of 700TB
upvoted 7 times

cookieMr 1 year, 11 months ago

Selected Answer: A

By ordering Snowball devices, the company can transfer the 700 TB of backup data from its data center to AWS. Once the data is transferred to S3, a lifecycle policy can be applied to automatically transition the files from the S3 Standard storage class to the cost-effective Amazon S3 Glacier Deep Archive storage class.

Option B would require continuous data transfer over the public internet, which could be time-consuming and costly given the large amount of data. It may also require significant bandwidth allocation.

Option C would involve additional costs for provisioning and maintaining the dedicated connection, which may not be necessary for a one-time data migration.

Option D could be a viable option, but it may incur additional costs for deploying and managing the DataSync agent.

Therefore, option A is the recommended choice as it provides a secure and efficient data transfer method using Snowball devices and allows for cost optimization through lifecycle policies by transitioning the data to S3 Glacier Deep Archive for long-term storage.
upvoted 4 times

arjundevops 2 years, 1 month ago

A is the correct answer.
even though they have 500mbps internetspeed, it will take around 130days to transfer the data from on premises to AWS

so they have only 1 option which is Snowball devices
upvoted 3 times

Paras043 2 years, 2 months ago

Selected Answer: A

A is the correct one
upvoted 2 times

CapJackSparrow 2 years, 2 months ago

Q: What is AWS Snowball Edge?

AWS Snowball Edge is an edge computing and data transfer device provided by the AWS Snowball service. It has on-board storage and compute power that provides select AWS services for use in edge locations. Snowball Edge comes in two options, Storage Optimized and Compute Optimized, to support local data processing and collection in disconnected environments such as ships, windmills, and remote factories. Learn more about its features here.

Q: What happened with the original 50 TB and 80 TB AWS Snowball devices?

The original Snowball devices were transitioned out of service and Snowball Edge Storage Optimized are now the primary devices used for data transfer.

Q: Can I still order the original Snowball 50 TB and 80 TB devices?

No. For data transfer needs now, please select the Snowball Edge Storage Optimized devices.
upvoted 2 times

vherman 2 years, 3 months ago

Selected Answer: A

Snowball
upvoted 1 times

KZM 2 years, 3 months ago

9 Snowball devices are needed to migrate the 700TB of data.
upvoted 3 times

KZM 2 years, 3 months ago

700TB of Data can not be transferred through a 500Mbps link within one month.

Total data that can be transferred in one month = bandwidth x time
= (500 Mbps / 8 bits per byte) x (30 days x 24 hours x 3600 seconds per hour)
= 648,000 GB or 648 TB
This is calculated theoretically with the maximum available situation. Due to a number of factors, the actual total transferred Data may be less than 645 TB.
upvoted 4 times

mandragon 2 years, 1 month ago

Good thinking. Agree with the solution. Only the calculation is wrong. It should give 162tb as a result
upvoted 4 times

Rudraman 2 years, 4 months ago

Snow ball Devices the answe is AAAAA.
upvoted 3 times

wmp7039 2 years, 4 months ago

A is incorrect as DC is an expensive option. Correct answer should be C as the company already has 500Mbps that can be used for data transfer. By consuming all the available internet bandwidth, data transfer will complete in 3 hours 6 mins - https://www.omnicalculator.com/other/data-transfer
upvoted 1 times

wmp7039 2 years, 4 months ago

Ignore please, miscalculated time to transfer, it will take 129 days and will breach the 1 month requirement. A is correct.
upvoted 6 times

kbaruu 2 years, 4 months ago

Selected Answer: A

A is correct
upvoted 2 times

EXAMTOPICS

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #216 *Topic 1*

A company has a serverless website with millions of objects in an Amazon S3 bucket. The company uses the S3 bucket as the origin for an Amazon CloudFront distribution. The company did not set encryption on the S3 bucket before the objects were loaded. A solutions architect needs to enable encryption for all existing objects and for all objects that are added to the S3 bucket in the future.

Which solution will meet these requirements with the LEAST amount of effort?

A. Create a new S3 bucket. Turn on the default encryption settings for the new S3 bucket. Download all existing objects to temporary local storage. Upload the objects to the new S3 bucket.

B. Turn on the default encryption settings for the S3 bucket. Use the S3 Inventory feature to create a .csv file that lists the unencrypted objects. Run an S3 Batch Operations job that uses the copy command to encrypt those objects. Most Voted

C. Create a new encryption key by using AWS Key Management Service (AWS KMS). Change the settings on the S3 bucket to use server-side encryption with AWS KMS managed encryption keys (SSE-KMS). Turn on versioning for the S3 bucket.

D. Navigate to Amazon S3 in the AWS Management Console. Browse the S3 bucket's objects. Sort by the encryption field. Select each unencrypted object. Use the Modify button to apply default encryption settings to every unencrypted object in the S3 bucket.

Correct Answer: *B*

Community vote distribution

B (91%) Other (9%)

Comments

Parsons Highly Voted 1 year, 10 months ago

Selected Answer: B

Step 1: S3 inventory to get object list
Step 2 (If needed): Use S3 Select to filter
Step 3: S3 object operations to encrypt the unencrypted objects.

On the going object use default encryption.

upvoted 17 times

Parsons 1 year, 10 months ago

Useful ref link: https://aws.amazon.com/blogs/storage/encrypting-objects-with-amazon-s3-batch-operations/
upvoted 10 times

cookieMr Highly Voted 1 year, 5 months ago

Selected Answer: B

By enabling default encryption settings on the S3, all newly added objects will be automatically encrypted. To encrypt the existing objects, the S3 Inventory feature can be used to generate a list of unencrypted objects. Then, an S3 Batch Operations job can be executed to copy those objects while applying encryption.

A. This solution involves creating a new S3 and manually downloading and uploading all existing objects. It requires significant effort and time to transfer millions of objects, making it a less efficient solution.

C. While enabling SSE with AWS KMS is a valid approach to encrypt objects in an S3, it does not address the requirement of encrypting existing objects. It only applies encryption to new objects added to the bucket.

D. Manually modifying each object in the S3 to apply default encryption settings is a labor-intensive and error-prone process. It would require individually selecting and modifying each unencrypted object, which is impractical for a large number of objects.
upvoted 15 times

iamroyalty_k Most Recent 4 months, 1 week ago

Selected Answer: B

Option B uses AWS tools like S3 Default Encryption, Inventory, and Batch Operations to enable encryption for both existing and future objects with minimal effort and automation, making it the best solution.

Why not the other options?
A. Create a new S3 bucket and migrate objects:
This involves downloading and re-uploading all objects, which is highly manual, time-consuming, and prone to errors. It also incurs additional data transfer costs.

C. Use SSE-KMS and turn on versioning:
While enabling SSE-KMS for future objects is valid, this does not encrypt the existing objects unless explicitly copied or rewritten, requiring additional manual effort.

D. Browse and modify objects manually:
Manually selecting and encrypting each object is impractical for a bucket with millions of objects, as it is labor-intensive and time-consuming.
upvoted 1 times

lofzee 6 months, 1 week ago

Selected Answer: B

to be fair all these options take a hell a lot of work to do but i think the least amount of effort is B.
https://aws.amazon.com/blogs/storage/encrypting-objects-with-amazon-s3-batch-operations/
upvoted 4 times

pentium75 11 months, 2 weeks ago

Selected Answer: B

A - Extreme amount of effort
B - Should work
C - SSE-KMS is not "least amount of effort" compared to SSE-S3; Turning versioning is not required to achieve the result but on the contrary, it will cause the non-encrypted files to remain as old versions even if you encrypt them in the future.
D - Even more effort as A
upvoted 2 times

foha2012 10 months, 3 weeks ago

B doesnt look like least amount of effort
upvoted 1 times

foha2012 10 months, 3 weeks ago

.csv for millions of objects ?? C looks simpler.
upvoted 2 times

CapJackSparrow 1 year, 8 months ago

Selected Answer: B

B...

https://catalog.us-east-1.prod.workshops.aws/workshops/05f16f1a-0bbf-45a7-a304-4fcd7fca3d1f/en-US/s3-track/module-2

You're welcome
upvoted 4 times

bdp123 1 year, 9 months ago

Selected Answer: B

Amazon S3 now configures default encryption on all existing unencrypted buckets to apply server-side encryption with S3 managed keys (SSE-S3) as the base level of encryption for new objects uploaded to these buckets. Objects that are already in an existing unencrypted bucket won't be automatically encrypted.
https://docs.aws.amazon.com/AmazonS3/latest/userguide/default-encryption-faq.html
upvoted 4 times

Yelizaveta 1 year, 9 months ago

Selected Answer: B

https://docs.aws.amazon.com/AmazonS3/latest/userguide/batch-ops-copy-example-bucket-key.html
upvoted 2 times

aakashkumar1999 1 year, 10 months ago

Selected Answer: B

B is the correct answer
upvoted 2 times

Val182 1 year, 10 months ago

Selected Answer: B

B 100%
https://spin.atomicobject.com/2020/09/15/aws-s3-encrypt-existing-objects/
upvoted 2 times

LuckyAro 1 year, 10 months ago

Selected Answer: A

Why is no one discussing A ? I think A can also achieve the required results. B is the most appropriate answer though.
upvoted 1 times

pentium75 11 months, 2 weeks ago

Downloading and uploading "millions of objects" is surely not "least amount of effort", thus does not meet the requirements.
upvoted 2 times

Training4aBetterLife 1 year, 10 months ago

Selected Answer: B

S3 provides a single control to automatically encrypt all new objects in a bucket with SSE-S3 or SSE-KMS. Unfortunately, these controls only affect new objects. If your bucket already contains millions of unencrypted objects, then turning on automatic encryption does not make your bucket secure as the unencrypted objects remain.

For S3 buckets with a large number of objects (millions to billions), use Amazon S3 Inventory to get a list of the unencrypted objects, and Amazon S3 Batch Operations to encrypt the large number of old, unencrypted files.
upvoted 4 times

Training4aBetterLife 1 year, 10 months ago

Versioning:

When you overwrite an S3 object, it results in a new object version in the bucket. However, this will not remove the old unencrypted versions of the object. If you do not delete the old version of your newly encrypted objects, you will be charged for the storage of both versions of the objects.

S3 Lifecycle

If you want to remove these unencrypted versions, use S3 Lifecycle to expire previous versions of objects. When you add a Lifecycle configuration to a bucket, the configuration rules apply to both existing objects and objects added later. C is missing this step, which I believe is what makes B the better choice. B includes the functionality of encrypting the old unencrypted objects via Batch Operations, whereas, Versioning does not address the old unencrypted objects.

upvoted 2 times

Training4aBetterLife 1 year, 10 months ago

S3 provides a single control to automatically encrypt all new objects in a bucket with SSE-S3 or SSE-KMS. Unfortunately, these controls only affect new objects. If your bucket already contains millions of unencrypted objects, then turning on automatic encryption does not make your bucket secure as the unencrypted objects remain.

For S3 buckets with a large number of objects (millions to billions), use Amazon S3 Inventory to get a list of the unencrypted objects, and Amazon S3 Batch Operations to encrypt the large number of old, unencrypted files.

upvoted 2 times

Training4aBetterLife 1 year, 10 months ago

Versioning:

When you overwrite an S3 object, it results in a new object version in the bucket. However, this will not remove the old unencrypted versions of the object. If you do not delete the old version of your newly encrypted objects, you will be charged for the storage of both versions of the objects.

S3 Lifecycle

If you want to remove these unencrypted versions, use S3 Lifecycle to expire previous versions of objects. When you add a Lifecycle configuration to a bucket, the configuration rules apply to both existing objects and objects added later. C is missing this step, which I believe is what makes B the better choice. B includes the functionality of encrypting the old unencrypted objects via Batch Operations, whereas, Versioning does not address the old unencrypted objects.

upvoted 1 times

Training4aBetterLife 1 year, 10 months ago

Please remove duplicate response as I was meaning to submit a voting comment.

upvoted 1 times

John_Zhuang 1 year, 10 months ago

Selected Answer: B

C is wrong. Even though you turn on the SSE-KMS with a new key, the existing objects are still yet to be encrypted. They still need to be manually encrypted by AWS batch

upvoted 3 times

pentium75 11 months, 2 weeks ago

And as in C you "turn on versioning", the old, unencrypted objects will be kept.

upvoted 2 times

LuckyAro 1 year, 10 months ago

Selected Answer: B

https://spin.atomicobject.com/2020/09/15/aws-s3-encrypt-existing-objects/

upvoted 2 times

Aninina 1 year, 10 months ago

Selected Answer: C

C is the answer

upvoted 2 times

pentium75 11 months, 2 weeks ago

Why? This does not include a step to encrypt existing objects, and by turning on versioning you will keep the unencrypted versions forever.

upvoted 1 times

techmb 1 year, 10 months ago

Selected Answer: B

Agree with Parsons

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #217 *Topic 1*

A company runs a global web application on Amazon EC2 instances behind an Application Load Balancer. The application stores data in Amazon Aurora. The company needs to create a disaster recovery solution and can tolerate up to 30 minutes of downtime and potential data loss. The solution does not need to handle the load when the primary infrastructure is healthy.

What should a solutions architect do to meet these requirements?

A. Deploy the application with the required infrastructure elements in place. Use Amazon Route 53 to configure active-passive failover. Create an Aurora Replica in a second AWS Region. **Most Voted**

B. Host a scaled-down deployment of the application in a second AWS Region. Use Amazon Route 53 to configure active-active failover. Create an Aurora Replica in the second Region.

C. Replicate the primary infrastructure in a second AWS Region. Use Amazon Route 53 to configure active-active failover. Create an Aurora database that is restored from the latest snapshot.

D. Back up data with AWS Backup. Use the backup to create the required infrastructure in a second AWS Region. Use Amazon Route 53 to configure active-passive failover. Create an Aurora second primary instance in the second Region.

Correct Answer: *A*

Community vote distribution

A (72%) D (28%)

Comments

Parsons Highly Voted 2 years, 4 months ago

Selected Answer: A

A is correct.
- "The solution does not need to handle the load when the primary infrastructure is healthy." => Should use Route 53 Active-Passive ==> Exclude B, C
- D is incorrect because "Create an Aurora second primary instance in the second Region.", we need to create an Aurora Replica enough.

upvoted 31 times

Parsons 2 years, 4 months ago

Ref link: https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/dns-failover-types.html

upvoted 6 times

diabloexodia Highly Voted 1 year, 10 months ago

Selected Answer: A

Anything that is not instant recovery is active - passive.
In active -passive we have :
1. Aws Backup(least op overhead) - RTO/RPO = hours
2. Pilot Light (Basic Infra is already deployed, but needs to be fully implemented) -RTO/RPO = 10's of minutes.
3. Warm Standby- (Basic infra + runs small loads (might need to add auto scaling) -RTO/RPO= minutes
4. (ACTIVE -ACTIVE) : Multi AZ option : instant

here we can tolerate 30 mins
hence B,D are incorrect. AWS backup is in hours, hence D is incorrect .
therefore A

upvoted 24 times

pentium75 1 year, 5 months ago

A does not create the infrastructure in the DR region though.

upvoted 2 times

bora4motion Most Recent 1 month ago

Selected Answer: A

A seems logic to me. D seems legit as well but it does not guarantee everything will be live in 30 minutes.

upvoted 1 times

FlyingHawk 6 months, 2 weeks ago

Selected Answer: A

The Aurora Global Database would be ideal for a production system requiring near-instantaneous failover, however, it was not in the options. Since the option D may not meet the 30-minute tolerance, I choose option A- Aurora Replica-based disaster recovery although replication is more designed for reading traffic, it can still be used for failover based on this doc https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/Aurora.Replication.html.

upvoted 2 times

ensbrvsnss 9 months, 2 weeks ago

Selected Answer: D

D is right

upvoted 1 times

jatric 11 months ago

Selected Answer: D

confused with A and D but D looks more promising when it says doesn't need to handle the load when primary infrastructure is healthy

upvoted 2 times

lofzee 1 year ago

Selected Answer: D

I went for D as the wording of A is weird....
D seems most plausible

upvoted 2 times

Jazz888 1 year ago

A
For those you are choosing D, I have a question for you. How do you guarantee the provisioning of resources will take less than 30 min through AWS Backup?

upvoted 2 times

ManikRoy 1 year, 1 month ago

By excluding other options you can choose A but this option is incomplete as it doesn't mention deploying/recovering the application in secondary region.

upvoted 1 times

MrPCarrot 1 year, 3 months ago

A is perfect - Active-Passive Failover: Use this failover configuration when you want a primary group of resources to be available the majority of the time and you want a secondary group of resources to be on standby in case all of the primary resources become unavailable.

upvoted 2 times

MrPCarrot 1 year, 3 months ago

A is perfect

upvoted 1 times

farnamjam 1 year, 4 months ago

Selected Answer: A

Here's why the other options aren't as suitable:

B. Active-active failover: Incur higher costs due to running both infrastructures simultaneously and introduces complexity in managing traffic distribution.
C. Restoring from snapshot: Could take longer than 30 minutes to recover, exceeding the company's downtime tolerance.
D. AWS Backup: Dependent on backup and restore times, potentially exceeding the 30-minute recovery window.

upvoted 2 times

pentium75 1 year, 5 months ago

Selected Answer: D

Not A - does not mention a second region for the infrastructure elements. Also, you cannot really "create an Aurora Replica in a second AWS Region", replicas must be in same region unless using Aurora Global Database (which is not mentioned)
Not B - would send half of the traffic to the DR region
Not C - this could send traffic to the DR instance even when the primary instance is healthy
D - the wording "Aurora second primary instance" is a bit strange, but still a "primary instance" is what we would need in the other region. We would still need to establish replication between the databases (like binlog), or restore a snapshot before failover, but in general this option could meet the 30 minute RTO/RPO requirement.

upvoted 5 times

lofzee 1 year ago

the question doesnt state Aurora MySQL but you can set up cross-region replicas for Aurora MySQL, but not PostgreSQL. the question only says Amazon Aurora, so its left a bit open as Aurora is either MySql or PostgreSQL. this is without using Global Database.
tbh i think this question and answers are well off... it is a dump after all.

upvoted 2 times

awsgeek75 1 year, 5 months ago

Selected Answer: D

I agreed with D as the requirements of 30 min downtime and potential data loss and no load consideration when primary instance is healthy. It makes D more feasible than A. Aurora-Replica is normally used for active-active failovers. Be frugal!

upvoted 4 times

Jeffab 1 year, 7 months ago

If this is the quality of the questions in exam, then we are all screwed! I don't think any options are correct. A proabably the most correct, but a big flaw. "Deploy the application with the required infrastructure elements in place." Deploy to where? Fair enough if you assume another region/AZ, but it's not stated and only Aurora replica is mentioned, not the Web/app servers etc.

upvoted 13 times

awsgeek75 1 year, 4 months ago

I really hope the language is better in the exams. Option A is like "do what it takes it to make the solution work".... well then by default it is right until the second part makes it wrong. smh!

upvoted 2 times

TariqKipkemei 1 year, 8 months ago

Selected Answer: A

'Can tolerate up to 30 minutes of downtime and potential data loss' rules out any option with 'active-active'. Leaves D and A. D is convoluted. Leaving A.

upvoted 6 times

upvoted 6 times

cookieMr 1 year, 11 months ago

Selected Answer: A

A. involves deploying the application and infrastructure elements in the primary Region. An Aurora Replica is created in a second Region to serve as the standby database. Route 53 is configured with active-passive failover, directing traffic to the primary Region by default. In the event of a disaster, Route 53 can automatically redirect traffic to the standby Region, minimizing downtime. Data loss may occur up to the point of the last replication to the standby Region, which can be within the defined tolerance of 30 minutes.

Option B, is not necessary in this case as the solution does not need to handle the load when the primary infrastructure is healthy, and it may involve higher complexity and costs.

Option C, may introduce additional complexity and potential data loss, as the standby database might not be up-to-date with the primary database.

Option D, may be suitable for backup and recovery scenarios but may not provide the required failover and downtime tolerance specified in the requirements.

upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #218 *Topic 1*

A company has a web server running on an Amazon EC2 instance in a public subnet with an Elastic IP address. The default security group is assigned to the EC2 instance. The default network ACL has been modified to block all traffic. A solutions architect needs to make the web server accessible from everywhere on port 443.

Which combination of steps will accomplish this task? (Choose two.)

A. Create a security group with a rule to allow TCP port 443 from source 0.0.0.0/0. **Most Voted**

B. Create a security group with a rule to allow TCP port 443 to destination 0.0.0.0/0.

C. Update the network ACL to allow TCP port 443 from source 0.0.0.0/0.

D. Update the network ACL to allow inbound/outbound TCP port 443 from source 0.0.0.0/0 and to destination 0.0.0.0/0.

E. Update the network ACL to allow inbound TCP port 443 from source 0.0.0.0/0 and outbound TCP port 32768-65535 to destination 0.0.0.0/0. **Most Voted**

Correct Answer: *AE*

Community vote distribution

AE (73%) E (13%) Other (14%)

Comments

Parsons Highly Voted 2 years, 4 months ago

Selected Answer: AE

A, E is perfect the combination. To be more precise, We should add outbound with "outbound TCP port 32768-65535 to destination 0.0.0.0/0." as an ephemeral port due to the stateless of NACL.

upvoted 20 times

oguzbeliren 1 year, 10 months ago

What is the main reason that you are using the TCP port 32768-65535> In the question, it doesn't ask you any requirement about it.

upvoted 5 times

MohammadTofic8787 1 year, 8 months ago

i Think AD because acl is stateless we must open the port outbound and inbound , in option c we only open 443 on inbound

upvoted 3 times

bora4motion 1 month ago

with d you only alow replies using port 443 which is wrong.
upvoted 1 times

MohammadTofic8787 1 year, 8 months ago

i Think AD because acl is stateless we must open the port outbound and inbound , in option E we only open 443 on inbound
upvoted 3 times

pentium75 Highly Voted 1 year, 5 months ago

Selected Answer: E

For me it's grammatically unclear whether "port 443" and "port 32768-65535" in answers D and E are referring to the source or destination ports of the outbound traffic. If source ports then it would be D. If destination ports (which seems more likely) then it's E.

"On Windows, the ephemeral port range is usually from 49152 to 65535.
On Linux, it is often from 32768 to 61000."

Thus 32768-65535 would cover both Windows and Linux.
upvoted 10 times

Omariox Most Recent 8 months ago

Selected Answer: AD

Option A: Creating a security group that allows inbound traffic on TCP port 443 from all sources (0.0.0.0/0) ensures that the web server can accept incoming HTTPS requests.

Option D: Updating the network ACL to allow inbound traffic on TCP port 443 from all sources (0.0.0.0/0) allows the requests to reach the EC2 instance. Additionally, it is necessary to allow outbound traffic on TCP port 443 to enable responses to clients, which is crucial for HTTPS communication.
upvoted 3 times

bora4motion 1 month ago

D is wrong: when the ec2 replies back it will use a random port not the same port (443) for the incoming traffic. AE.
upvoted 1 times

srinibas.velumuri 8 months, 3 weeks ago

Higher priority NACL to allow inbound and outbound traffic on 443 with take the precedence over default blocked NACL
upvoted 1 times

ChinthaGurumurthi 10 months, 2 weeks ago

Selected Answer: AD

AD
How can E be the answer. How can we assure that the port range is definitely from the given port range in the option E?
upvoted 1 times

lofzee 1 year ago

Selected Answer: AE

Security group only needs inbound rules.
ACL needs inbound and outbound.. Outbound traffic is going to be dynamic ports. Answer is A and E
upvoted 4 times

sidharthwader 1 year, 3 months ago

AE

Security group is a stateful resource and can understand to allow traffic from source 0.0.0.0/0 with port 443 but ACL is stateless so traffic that is allowed inside the network we must configure the same to go outside the network as well.
upvoted 2 times

awsgeek75 1 year, 5 months ago

Selected Answer: AE

https://docs.aws.amazon.com/vpc/latest/userguide/vpc-network-acls.html#nacl-basics
"NACLs are stateless, which means that information about previously sent or received traffic is not saved. If, for example, you create a NACL rule to allow specific inbound traffic to a subnet, responses to that traffic are not automatically allowed. This is in contrast to how security groups work. Security groups are stateful, which means that information about previously sent or received traffic is saved. If, for example, a security group allows inbound traffic to an EC2 instance, responses are automatically allowed regardless of outbound security group rules."
A fulfils the security group requirement
E is the only option that explicitly covers outbound traffic and ports.
D covers outbound destination but given that all traffic is blocked (as per the question) this won't work

upvoted 5 times

[Removed] 1 year, 6 months ago

Selected Answer: AC

For typical web server scenarios, such as serving content over HTTPS (port 443), you generally do not need to explicitly open outbound ports in the network ACL (NACL) for the return traffic.

upvoted 1 times

pentium75 1 year, 5 months ago

But NACLs are stateless."The default network ACL has been modified to block all traffic"; if you don't allow any outbound traffic then the web server won't be able to reply to clients.

upvoted 2 times

TariqKipkemei 1 year, 8 months ago

Selected Answer: AE

ACL is stateless. you have to define both inbound and outbound rules.

upvoted 3 times

MohammadTofic8787 1 year, 8 months ago

i Think AD because acl is stateless we must open the port outbound and inbound , in option c we only open 443 on inbound

upvoted 2 times

MohammadTofic8787 1 year, 8 months ago

i Think AD because acl is stateless we must open the port outbound and inbound , in option D we only open 443 on inbound

upvoted 1 times

MohammadTofic8787 1 year, 8 months ago

please admin delete this , sorry

upvoted 1 times

MohammadTofic8787 1 year, 8 months ago

please admin delete this , sorry

upvoted 1 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: AE

A, E is perfect the combination. To be more precise, We should add outbound with "outbound TCP port 32768-65535 to destination 0.0.0.0/0." as an ephemeral port due to the stateless of NACL.

upvoted 4 times

beginnercloud 1 year, 9 months ago

Selected Answer: AE

AE is the best answer here, but in reality, E is not good enough. Here, it says that the client chooses the ephemeral port, and it can start from 1024. Only Linux clients have the range starting at 32768 https://docs.aws.amazon.com/vpc/latest/userguide/vpc-network-acls.html#nacl-ephemeral-ports Unless the destination advertises the ephemeral ports, which I don't think is the case

upvoted 2 times

pentium75 1 year, 5 months ago

"On Windows, the ephemeral port range is usually from 49152 to 65535.
On Linux, it is often from 32768 to 61000."
Combined: 32768 - 65535 ...

upvoted 3 times

Thornessen 1 year, 10 months ago

Selected Answer: AE

AE is the best answer here, but in reality, E is not good enough.

Here, it says that the client chooses the ephemeral port, and it can start from 1024. Only Linux clients have the range starting at 32768 https://docs.aws.amazon.com/vpc/latest/userguide/vpc-network-acls.html#nacl-ephemeral-ports

Unless the destination advertises the ephemeral ports, which I don't think is the case

upvoted 3 times

Abrar2022 2 years ago

32768-65535 ports Allows outbound IPv4 responses to clients on the internet (for example, serving webpages to people visiting the web servers in the subnet).

upvoted 2 times

WherecanIstart 2 years, 2 months ago

Selected Answer: AE

NACL blocks outgoing traffic since it is infact stateless..Option E allows outbound traffic from ephemeral ports going outside of the VPC back to the web.

upvoted 3 times

Brak 2 years, 3 months ago

It can't be C, since the current NACL blocks all traffic, including outbound. Need to allow outbound traffic through the NACL. But E is a bad answer, since ephemeral ports start at 1024, not 32768.

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #219 *Topic 1*

A company's application is having performance issues. The application is stateful and needs to complete in-memory tasks on Amazon EC2 instances. The company used AWS CloudFormation to deploy infrastructure and used the M5 EC2 instance family. As traffic increased, the application performance degraded. Users are reporting delays when the users attempt to access the application.

Which solution will resolve these issues in the MOST operationally efficient way?

A. Replace the EC2 instances with T3 EC2 instances that run in an Auto Scaling group. Make the changes by using the AWS Management Console.

B. Modify the CloudFormation templates to run the EC2 instances in an Auto Scaling group. Increase the desired capacity and the maximum capacity of the Auto Scaling group manually when an increase is necessary.

C. Modify the CloudFormation templates. Replace the EC2 instances with R5 EC2 instances. Use Amazon CloudWatch built-in EC2 memory metrics to track the application performance for future capacity planning.

D. Modify the CloudFormation templates. Replace the EC2 instances with R5 EC2 instances. Deploy the Amazon CloudWatch agent on the EC2 instances to generate custom application latency metrics for future capacity planning.
Most Voted

Correct Answer: *D*

Community vote distribution

D (94%) Other

Comments

Parsons Highly Voted 2 years, 4 months ago

Selected Answer: D

D is the correct answer.

"in-memory tasks" => need the "R" EC2 instance type to archive memory optimization. So we are concerned about C & D.
Because EC2 instances don't have built-in memory metrics to CW by default. As a result, we have to install the CW agent to archive the purpose.

upvoted 39 times

Babba Highly Voted 2 years, 4 months ago

Selected Answer: D

Selected Answer: D

It's D, EC2 do not provide by default memory metrics to CloudWatch and require the CloudWatch Agent to be installed on the monitored instances : https://aws.amazon.com/premiumsupport/knowledge-center/cloudwatch-memory-metrics-ec2/

upvoted 14 times

surajkrishnamurthy Most Recent 3 months, 3 weeks ago

Selected Answer: D

Answer is most likely "D" because of the following ...
- T3 EC2 instances > low cost burstable general purpose instance type that provide a baseline level of CPU performance with the ability to burst CPU usage
- M5 EC2 instance > M5 instances offer a balance of compute, memory, and networking resources for a broad range of workload.
- R5 EC2 instances > Amazon EC2 R5 instances are the next generation of memory optimized instances for the Amazon Elastic Compute Cloud. R5 instances are well suited for memory intensive applications such as high-performance databases etc

M5 EC2 instance is already in use & having performance issues so there is no point in provisioning a lower cost , general purpose instance type like T3 EC2.
So as per the given Answer option choice the best one is Answer D

upvoted 1 times

jatric 11 months ago

Selected Answer: A

how future capacity planning and just do verticall scalling will improve the performance. Question doesn't specify if these EC2 are behind auto scalling so it means they are not. out of all A seems more close to the solution

upvoted 3 times

a7md0 11 months, 3 weeks ago

Selected Answer: B

B will reduce operational overhead and better solution than keep changing the family. Also, I don't think the exam will require you to remember instance families like M5 and R5

upvoted 2 times

lofzee 1 year ago

Selected Answer: D

D .....

upvoted 2 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: D

R5 instances are better optimized for the in-memory workload than M5.
Auto Scaling alone doesn't handle stateful applications well, manual capacity adjustments would still be needed.
Custom latency metrics give better visibility than built-in metrics for capacity planning.

upvoted 7 times

cookieMr 1 year, 11 months ago

Selected Answer: D

By replacing the M5 instances with R5 instances, which are optimized for memory-intensive workloads, the application can benefit from increased memory capacity and performance.

In addition, deploying the CloudWatch agent on the EC2 instances allows for the generation of custom application latency metrics, which can provide valuable insights into the application's performance.

This solution addresses the performance issues efficiently by leveraging the appropriate instance types and collecting custom application metrics for better monitoring and future capacity planning.

A. Replacing with T3 instances may not provide enough memory capacity for in-memory tasks.

B. Manually increasing the capacity of the ASG does not directly address the performance issues.

C. Relying solely on built-in EC2 memory metrics may not provide enough granularity for optimizing in-memory tasks.

The most efficient solution is to modify the CloudFormation templates, replace with R5 instances, and deploy the CloudWatch agent for custom metrics.

upvoted 6 times

upvoted 6 times

Bmarodi 2 years ago

Selected Answer: D

Option D is the correct answer.

upvoted 2 times

BABU97 2 years, 2 months ago

will go for C

upvoted 1 times

Aninina 2 years, 4 months ago

Selected Answer: D

Would go with D

upvoted 2 times

mhmt4438 2 years, 4 months ago

Selected Answer: D

ı think D

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #220 *Topic 1*

A solutions architect is designing a new API using Amazon API Gateway that will receive requests from users. The volume of requests is highly variable; several hours can pass without receiving a single request. The data processing will take place asynchronously, but should be completed within a few seconds after a request is made.

Which compute service should the solutions architect have the API invoke to deliver the requirements at the lowest cost?

A. An AWS Glue job

B. An AWS Lambda function **Most Voted**

C. A containerized service hosted in Amazon Elastic Kubernetes Service (Amazon EKS)

D. A containerized service hosted in Amazon ECS with Amazon EC2

Correct Answer: *B*

Community vote distribution

B (97%) D

Comments

Parsons Highly Voted 2 years, 4 months ago

Selected Answer: B

B is the correct answer.
API Gateway + Lambda is the perfect solution for modern applications with serverless architecture.

upvoted 12 times

cookieMr Highly Voted 1 year, 11 months ago

Selected Answer: B

Lambda is a serverless compute service that can be triggered by API Gateway to process requests asynchronously. It automatically scales based on the incoming request volume and allows for cost optimization by charging only for the actual compute time used to process the requests.

A. Glue is a fully managed ETL service. It is designed for data processing and transformation tasks rather than serving API requests. It may not be suitable for handling variable request volumes and delivering responses within a few seconds.

C. While EKS provides scalability and flexibility, it may introduce additional complexity and overhead for managing and scaling the infrastructure for handling variable API request volumes.

D. Similar to the previous option, using ECS with EC2 would require additional effort for infrastructure management and scaling, which may not be necessary for handling intermittent and variable API request volumes.

upvoted 6 times

a7md0 Most Recent 11 months, 3 weeks ago

Selected Answer: B

Screaming Lambda

upvoted 4 times

TariqKipkemei 1 year, 8 months ago

Selected Answer: B

data processing should be completed within a few seconds = An AWS Lambda function

upvoted 2 times

JA2018 6 months, 3 weeks ago

definitely well within Lambda's maximum runtime limit of 15 minutes.

upvoted 1 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: B

B. An AWS Lambda function

upvoted 2 times

ukivanlamlpi 1 year, 9 months ago

Selected Answer: D

lambda is expensive than running ECS on EC2

upvoted 1 times

pentium75 1 year, 5 months ago

"Several hours can pass without receiving a single request", during which Lambda costs 0.00.

upvoted 5 times

Undisputed 1 year, 10 months ago

Selected Answer: B

Lambda all the way.

upvoted 2 times

Bmarodi 2 years ago

Selected Answer: B

Option B meets the requirements.

upvoted 2 times

Aninina 2 years, 4 months ago

Selected Answer: B

Lambda !

upvoted 4 times

mhmt4438 2 years, 4 months ago

Selected Answer: B

https://www.examtopics.com/discussions/amazon/view/43780-exam-aws-certified-solutions-architect-associate-saa-c02/

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #221 *Topic 1*

A company runs an application on a group of Amazon Linux EC2 instances. For compliance reasons, the company must retain all application log files for 7 years. The log files will be analyzed by a reporting tool that must be able to access all the files concurrently.

Which storage solution meets these requirements MOST cost-effectively?

A. Amazon Elastic Block Store (Amazon EBS)

B. Amazon Elastic File System (Amazon EFS)

C. Amazon EC2 instance store

D. Amazon S3 Most Voted

Correct Answer: *D*

Community vote distribution

D (100%)

Comments

cookieMr Highly Voted 1 year, 11 months ago

Selected Answer: D

A. EBS provides block-level storage volumes for use with EC2 instances. While it offers durability and persistence, it is not the most cost-effective solution for long-term retention of log files. Additionally, it does not provide concurrent access to the files, which is a requirement in this scenario.

B. EFS is a scalable file storage service that can be mounted on multiple EC2 instances concurrently. While it provides concurrent access to files, it may not be the most cost-effective option for long-term retention due to its higher pricing compared to S3.

C. The instance store is a temporary storage option that is physically attached to the EC2 instance. It does not provide the durability and long-term retention required for compliance purposes. Additionally, the instance store is not accessible outside of the specific EC2 instance it is attached to, so concurrent access by the reporting tool would not be possible.

Therefore, considering the requirements for long-term retention, concurrent access, and cost-effectiveness, S3 is the most suitable and cost-effective storage solution.

upvoted 14 times

satyaammm Most Recent 3 months, 2 weeks ago

Selected Answer: D

EFS files cannot be stored for more than 365 days even with lifecycle policies. S3 is the most suitable option here.
upvoted 1 times

Ruffyit 1 year, 6 months ago

A. EBS provides block-level storage volumes for use with EC2 instances. While it offers durability and persistence, it is not the most cost-effective solution for long-term retention of log files. Additionally, it does not provide concurrent access to the files, which is a requirement in this scenario.

B. EFS is a scalable file storage service that can be mounted on multiple EC2 instances concurrently. While it provides concurrent access to files, it may not be the most cost-effective option for long-term retention due to its higher pricing compared to S3.

C. The instance store is a temporary storage option that is physically attached to the EC2 instance. It does not provide the durability and long-term retention required for compliance purposes. Additionally, the instance store is not accessible outside of the specific EC2 instance it is attached to, so concurrent access by the reporting tool would not be possible.
upvoted 3 times

Chiquitabandita 1 year, 8 months ago

this sounds like an expensive solution but if necessary then S3 would be the best
upvoted 2 times

TariqKipkemei 1 year, 8 months ago

most cost effective = Amazon S3
upvoted 2 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: D

D. Amazon S3
upvoted 2 times

kapit 1 year, 11 months ago

s3<efs<ebs
upvoted 2 times

Iconique 1 year, 8 months ago

actually S3 < EBS < EFS, but for EBS you need to pay for the underlying provisioned GB.
If you compare 1 GB then S3 < EBS < EFS but if you have 100GB storage for EBS than EBS is more expensive.
upvoted 2 times

mattcl 1 year, 12 months ago

"The log files will be analyzed by a reporting tool that must be able to access all the files concurrently" , so you need to access concurrently to get the logs. So is EFS. Letter B
upvoted 1 times

northyork 1 year, 12 months ago

https://aws.amazon.com/efs/faq/
EFS is a file storage service for use with Amazon compute (EC2, containers, serverless) and on-premises servers. EFS provides a file system interface, file system access semantics (such as strong consistency and file locking), and concurrently accessible storage for up to thousands of EC2 instances.
upvoted 1 times

JA2018 6 months, 3 weeks ago

from: https://docs.aws.amazon.com/AmazonS3/latest/userguide/optimizing-performance.html

Your applications can easily achieve thousands of transactions per second in request performance when uploading and retrieving storage from Amazon S3. Amazon S3 automatically scales to high request rates. For example, your application can achieve at least

*** 3,500 PUT/COPY/POST/DELETE requests per second *** or

*** 5,500 GET/HEAD requests per second *** per partitioned Amazon S3 prefix.
upvoted 1 times

alexandercamachop 2 years ago

Selected Answer: D

Whenever we see long time storage and no special requirements that needs EFS or FSx, then S3 is the way.
upvoted 4 times

elearningtakai 2 years, 2 months ago

Selected Answer: D

To meet the requirements of retaining application log files for 7 years and allowing concurrent access by a reporting tool, while also being cost-effective, the recommended storage solution would be D: Amazon S3.
upvoted 3 times

osmk 2 years, 2 months ago

ddddddddddddddddd
upvoted 3 times

udo2020 2 years, 2 months ago

What about the keyword "concurrently"? Doesn't this mean EFS?
upvoted 3 times

Aninina 2 years, 4 months ago

Selected Answer: D

Cost Effective: S3
upvoted 3 times

Parsons 2 years, 4 months ago

Selected Answer: D

S3 is enough with the lowest cost perspective.
upvoted 2 times

mhmt4438 2 years, 4 months ago

Selected Answer: D

https://www.examtopics.com/discussions/amazon/view/22182-exam-aws-certified-solutions-architect-associate-saa-c02/
upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #222 *Topic 1*

A company has hired an external vendor to perform work in the company's AWS account. The vendor uses an automated tool that is hosted in an AWS account that the vendor owns. The vendor does not have IAM access to the company's AWS account.

How should a solutions architect grant this access to the vendor?

A. Create an IAM role in the company's account to delegate access to the vendor's IAM role. Attach the appropriate IAM policies to the role for the permissions that the vendor requires. **Most Voted**

B. Create an IAM user in the company's account with a password that meets the password complexity requirements. Attach the appropriate IAM policies to the user for the permissions that the vendor requires.

C. Create an IAM group in the company's account. Add the tool's IAM user from the vendor account to the group. Attach the appropriate IAM policies to the group for the permissions that the vendor requires.

D. Create a new identity provider by choosing "AWS account" as the provider type in the IAM console. Supply the vendor's AWS account ID and user name. Attach the appropriate IAM policies to the new provider for the permissions that the vendor requires.

Correct Answer: *A*

Community vote distribution

A (91%) Other (9%)

Comments

cookieMr Highly Voted 1 year, 5 months ago

By creating an IAM role and delegating access to the vendor's IAM role, you establish a trust relationship between accounts. This allows the vendor's automated tool to assume the role in the company's account and access the necessary resources.

By attaching the appropriate IAM policies to the role, you can define the precise permissions that the vendor requires for their tool to perform its tasks. This ensures that the vendor has the necessary access without granting them direct IAM access to the company's account.

B is incorrect because creating an IAM user with a password would require sharing the credentials with the vendor, which is not recommended for security reasons.

C is incorrect because adding the vendor's IAM user to an IAM group in the company's account would not provide a direct and controlled way to delegate access to the vendor's tool.

D is incorrect because creating a new identity provider for the vendor's AWS account would not provide a straightforward way to delegate access to the vendor's tool. Identity providers are typically used for federated access using external identity systems.

upvoted 11 times

mp165 Highly Voted 1 year, 10 months ago

Selected Answer: A

A is proper

https://docs.aws.amazon.com/IAM/latest/UserGuide/id_roles_common-scenarios_third-party.html

upvoted 10 times

satyaammm Most Recent 3 months, 2 weeks ago

Selected Answer: A

IAM roles are the most suitable here.

upvoted 1 times

Ruffyit 1 year ago

Create an IAM role in the company's account to delegate access to the vendor's IAM role. Attach the appropriate IAM policies to the role for the permissions that the vendor requires

upvoted 2 times

TariqKipkemei 1 year, 2 months ago

Selected Answer: A

Create an IAM role in the company's account to delegate access to the vendor's IAM role. Attach the appropriate IAM policies to the role for the permissions that the vendor requires

upvoted 2 times

Guru4Cloud 1 year, 2 months ago

Selected Answer: A

A. Create an IAM role in the company's account to delegate access to the vendor's IAM role. Attach the appropriate IAM policies to the role for the permissions that the vendor requires.

upvoted 2 times

teja54 1 year, 6 months ago

Selected Answer: C

....

upvoted 1 times

Bmarodi 1 year, 6 months ago

Selected Answer: A

Option A fulfill the requirements.

upvoted 2 times

Aninina 1 year, 10 months ago

Selected Answer: A

IAM role is the answer

upvoted 2 times

techhb 1 year, 10 months ago

Selected Answer: A

A is correct answer.

upvoted 2 times

kbaruu 1 year, 10 months ago

Selected Answer: A

https://docs.aws.amazon.com/IAM/latest/UserGuide/id_roles_common-scenarios_third-party.html

upvoted 3 times

upvoted 3 times

venice1234 1 year, 10 months ago

Selected Answer: A

https://docs.aws.amazon.com/IAM/latest/UserGuide/id_roles_create_for-user_externalid.html

upvoted 3 times

Parsons 1 year, 10 months ago

Selected Answer: A

A is the correct answer.

upvoted 4 times

Babba 1 year, 10 months ago

Selected Answer: D

My guess is D: https://docs.aws.amazon.com/IAM/latest/UserGuide/id_roles_common-scenarios_third-party.html

upvoted 2 times

pentium75 11 months, 2 weeks ago

But your link describes A, not D.

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #223 *Topic 1*

A company has deployed a Java Spring Boot application as a pod that runs on Amazon Elastic Kubernetes Service (Amazon EKS) in private subnets. The application needs to write data to an Amazon DynamoDB table. A solutions architect must ensure that the application can interact with the DynamoDB table without exposing traffic to the internet.

Which combination of steps should the solutions architect take to accomplish this goal? (Choose two.)

A. Attach an IAM role that has sufficient privileges to the EKS pod. **Most Voted**

B. Attach an IAM user that has sufficient privileges to the EKS pod.

C. Allow outbound connectivity to the DynamoDB table through the private subnets' network ACLs.

D. Create a VPC endpoint for DynamoDB. **Most Voted**

E. Embed the access keys in the Java Spring Boot code.

Correct Answer: *AD*

Community vote distribution

AD (100%)

Comments

iamroyalty_k 4 months, 1 week ago

Selected Answer: AD

A and D are the best choices because they ensure secure authentication with IAM roles and private connectivity to DynamoDB via a VPC endpoint.

B. Attach an IAM user that has sufficient privileges to the EKS pod:
IAM users are intended for individuals, not applications.

C. Allow outbound connectivity to the DynamoDB table through the private subnets' network ACLs:
While this ensures outbound traffic can reach DynamoDB, it doesn't eliminate the need for internet access unless a VPC endpoint is used.

E. Embed the access keys in the Java Spring Boot code:
Hardcoding access keys in the application code is a security risk.

upvoted 2 times

Burrito69 8 months, 2 weeks ago

After seeing D, I didn't even look at option E. its AD correct
upvoted 2 times

awsgeek75 10 months, 3 weeks ago

Selected Answer: AD

B: Wrong, cannot be a user for EKS
C: Not possible as NACL need destination CIDR/ports etc. This is not correct way to connect to DynamoDB
E: Not secure
AD is correct because you need roles for allowing service permissions and accessing DynamoDB with VPC endpoint is the correct way
upvoted 3 times

Ruffyit 1 year ago

The application needs to write data to an Amazon DynamoDB table = Attach an IAM role that has write privileges to the EKS pod
Without exposing traffic to the internet = VPC endpoint for DynamoDB
upvoted 2 times

TariqKipkemei 1 year, 2 months ago

Selected Answer: AD

The application needs to write data to an Amazon DynamoDB table = Attach an IAM role that has write privileges to the EKS pod
Without exposing traffic to the internet = VPC endpoint for DynamoDB
upvoted 4 times

Guru4Cloud 1 year, 2 months ago

Selected Answer: AD

A. By attaching an IAM role to the EKS pod, you can grant the necessary permissions for the pod to access DynamoDB. The IAM role should have appropriate policies allowing access to the DynamoDB table.

D. Creating a VPC endpoint for DynamoDB allows the EKS pod to access DynamoDB privately within the VPC, without the need for internet connectivity. The VPC endpoint provides a direct and secure connection to DynamoDB, eliminating the need for traffic to flow over the internet.
upvoted 3 times

cookieMr 1 year, 5 months ago

Selected Answer: AD

A. By attaching an IAM role to the EKS pod, you can grant the necessary permissions for the pod to access DynamoDB. The IAM role should have appropriate policies allowing access to the DynamoDB table.

D. Creating a VPC endpoint for DynamoDB allows the EKS pod to access DynamoDB privately within the VPC, without the need for internet connectivity. The VPC endpoint provides a direct and secure connection to DynamoDB, eliminating the need for traffic to flow over the internet.

B is incorrect because attaching an IAM user to the pod is not a recommended approach. IAM users are meant for accessing AWS services through the AWS Management Console or AP.

C is incorrect because configuring outbound connectivity through network ACLs would not provide a secure and direct connection to DynamoDB.

E is incorrect because embedding access keys in the code is not a recommended security practice. It can lead to potential security vulnerabilities. It is better to use IAM roles or other secure mechanisms for providing access to AWS services.
upvoted 3 times

Bmarodi 1 year, 6 months ago

Selected Answer: AD

A & D options fulfill the requirements.
upvoted 2 times

LuckyAro 1 year, 10 months ago

Selected Answer: AD

Definitely

upvoted 2 times

Aninina 1 year, 10 months ago

Selected Answer: AD

A D are the correct options

upvoted 2 times

venice1234 1 year, 10 months ago

Selected Answer: AD

https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/vpc-endpoints-dynamodb.html
https://aws.amazon.com/about-aws/whats-new/2019/09/amazon-eks-adds-support-to-assign-iam-permissions-to-kubernetes-service-accounts/

upvoted 3 times

Parsons 1 year, 10 months ago

Selected Answer: AD

A, D is the correct answer.

upvoted 3 times

mhmt4438 1 year, 10 months ago

Selected Answer: AD

The correct answer is A,D

upvoted 2 times

EXAMTOPICS

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #224 *Topic 1*

A company recently migrated its web application to AWS by rehosting the application on Amazon EC2 instances in a single AWS Region. The company wants to redesign its application architecture to be highly available and fault tolerant. Traffic must reach all running EC2 instances randomly.

Which combination of steps should the company take to meet these requirements? (Choose two.)

A. Create an Amazon Route 53 failover routing policy.

B. Create an Amazon Route 53 weighted routing policy.

C. Create an Amazon Route 53 multivalue answer routing policy. **Most Voted**

D. Launch three EC2 instances: two instances in one Availability Zone and one instance in another Availability Zone.

E. Launch four EC2 instances: two instances in one Availability Zone and two instances in another Availability Zone. **Most Voted**

Correct Answer: *CE*

Community vote distribution

CE (76%) BE (24%)

Comments

cookieMr Highly Voted 1 year, 11 months ago

Selected Answer: CE

C. A multivalue answer routing policy in Route 53 allows you to configure multiple values for a DNS record, and Route 53 responds to DNS queries with multiple random values. This enables the distribution of traffic randomly among the available EC2 instances.

E. By launching EC2 instances in different AZs, you achieve high availability and fault tolerance. Launching four instances (two in each AZ) ensures that there are enough resources to handle the traffic load and maintain the desired level of availability.

A. Failover routing is designed to direct traffic to a backup resource or secondary location only when the primary resource or location is unavailable.

B. Although a weighted routing policy allows you to distribute traffic across multiple EC2 instances, it does not ensure random distribution.

D. While launching instances in multiple AZs is important for fault tolerance, having only three instances does not provide an

even distribution of traffic. With only three instances, the traffic may not be evenly distributed, potentially leading to imbalanced resource utilization.
upvoted 20 times

Steve_4542636 Highly Voted 2 years, 3 months ago

Selected Answer: BE

I went back and rewatched the lectures from Udemy on Weighted and Multi-Value. The lecturer said that Multi-value is *not* as substitute for ELB and he stated that DNS load balancing is a good use case for Weighted routing policies
upvoted 9 times

smartegnine 1 year, 12 months ago

Weighted routing based on weight assigned, it can not do randomly choose, please see last sentence of the question choose randomly.
upvoted 9 times

foha2012 1 year, 4 months ago

what about 50 50 weighted ?
upvoted 2 times

satyaammm Most Recent 3 months, 2 weeks ago

Selected Answer: CE

Multi value routing will offer random distribution of traffic and 4 EC2 instances in different regions will offer HA
upvoted 1 times

iamroyalty_k 4 months, 1 week ago

Selected Answer: CE

To achieve high availability, fault tolerance, and random traffic distribution, the company should use Route 53 multivalue answer routing and launch four EC2 instances across two Availability Zones.

A. Failover routing policy: This is for active-passive setups, not for distributing traffic randomly.
B. Weighted routing policy: This allows control over traffic distribution proportions but doesn't ensure randomness or health checks.
D. Three EC2 instances in unbalanced zones: This setup lacks symmetry, reducing fault tolerance if the AZ with two instances fails.
upvoted 2 times

0de7d1b 6 months ago

Selected Answer: BE

By combining these two strategies, the company can achieve high availability and fault tolerance:

Weighted Routing Policy: This Route 53 policy distributes traffic across multiple endpoints based on weights assigned to each endpoint. By assigning equal weights to all EC2 instances, traffic can be distributed evenly across them.
Multiple Availability Zones: Deploying EC2 instances across multiple Availability Zones ensures that the application can continue to function even if one Availability Zone experiences an outage.https://www.examtopics.com/exams/amazon/aws-certified-solutions-architect-associate-saa-c03/view/23/#
upvoted 1 times

bignatov 8 months, 3 weeks ago

Selected Answer: CE

C and E are the correct answers.
upvoted 2 times

rohitph 1 year ago

Selected Answer: CE

CE is correct
upvoted 1 times

bujuman 1 year, 4 months ago

Selected Answer: CE

CE seems good to me due to " highly available and fault tolerant" and following explanation:
https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/routing-policy-multivalue.html
upvoted 2 times

bujuman 1 year, 4 months ago

Specifically with this requirement : "Traffic must reach all running EC2 instances randomly"
upvoted 2 times

awsgeek75 1 year, 4 months ago

Selected Answer: CE

E: For HA
C: Random routing can only be created with multivalue answer routing policy.

https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/routing-policy-multivalue.html

"To route traffic approximately randomly to multiple resources, such as web servers, you create one multivalue answer record for each resource and, optionally, associate a Route 53 health check with each record. "
upvoted 4 times

Ruffyit 1 year, 6 months ago

C. A multivalue answer routing policy in Route 53 allows you to configure multiple values for a DNS record, and Route 53 responds to DNS queries with multiple random values. This enables the distribution of traffic randomly among the available EC2 instances.

E. By launching EC2 instances in different AZs, you achieve high availability and fault tolerance. Launching four instances (two in each AZ) ensures that there are enough resources to handle the traffic load and maintain the desired level of availability.

A. Failover routing is designed to direct traffic to a backup resource or secondary location only when the primary resource or location is unavailable.

B. Although a weighted routing policy allows you to distribute traffic across multiple EC2 instances, it does not ensure random distribution.
upvoted 2 times

mohamoha 1 year, 7 months ago

First I thought it was weighted but after research C is the correct answer :
https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/routing-policy.html
upvoted 2 times

daniel33 1 year, 8 months ago

Selected Answer: CE

Multivalue routing can do random load balancing according to the AWS website:

To route traffic approximately randomly to multiple resources, such as web servers, you create one multivalue answer record for each resource and, optionally, associate a Route 53 health check with each record.

https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/routing-policy-multivalue.html
upvoted 5 times

Tom123456ac 1 year, 8 months ago

This questions is so wired , 3 instances nothing wrong with it
upvoted 1 times

pentium75 1 year, 5 months ago

I guess it's the "highly available AND fault tolerant" requirement. If AZ 1 fails, you have only a single server left in the other region.
upvoted 3 times

Techi47 1 year, 8 months ago

Option CE Correct:
To route traffic roughly and randomly to multiple resources, such as web servers, you create a multi-value response record for each resource and optionally associate a Route 53 health check with each record.

https://disaster-recovery.workshop.aws/en/services/networking/route53/routing-policies/routing-multiple-answer.html
upvoted 2 times

kwang312 1 year, 8 months ago

Selected Answer: CE

CE is correct
upvoted 2 times

TariqKipkemei 1 year, 8 months ago

Selected Answer: CE

Highly available and fault tolerant = two instances in two AZs
Route traffic randomly = Amazon Route 53 multivalue answer routing policy
upvoted 2 times

LazyTs 1 year, 9 months ago

Selected Answer: CE

Multivalue answer routing policy – Use when you want Route 53 to respond to DNS queries with up to eight healthy records selected at random. You can use multivalue answer routing to create records in a private hosted zone.

Weighted routing policy – Use to route traffic to multiple resources in proportions that you specify. You can use weighted routing to create records in a private hosted zone.
upvoted 2 times

foha2012 1 year, 4 months ago

what about 50 50 weighted routing ?
upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #225 *Topic 1*

A media company collects and analyzes user activity data on premises. The company wants to migrate this capability to AWS. The user activity data store will continue to grow and will be petabytes in size. The company needs to build a highly available data ingestion solution that facilitates on-demand analytics of existing data and new data with SQL.

Which solution will meet these requirements with the LEAST operational overhead?

A. Send activity data to an Amazon Kinesis data stream. Configure the stream to deliver the data to an Amazon S3 bucket.

B. Send activity data to an Amazon Kinesis Data Firehose delivery stream. Configure the stream to deliver the data to an Amazon Redshift cluster. Most Voted

C. Place activity data in an Amazon S3 bucket. Configure Amazon S3 to run an AWS Lambda function on the data as the data arrives in the S3 bucket.

D. Create an ingestion service on Amazon EC2 instances that are spread across multiple Availability Zones. Configure the service to forward data to an Amazon RDS Multi-AZ database.

Correct Answer: *B*

Community vote distribution

B (96%) A

Comments

beginnercloud Highly Voted 1 year, 9 months ago

Selected Answer: B

Petabyte scale- Redshift

upvoted 13 times

Berny Highly Voted 2 years, 4 months ago

Selected Answer: B

Data ingestion through Kinesis data streams will require manual intervention to provide more shards as data size grows. Kinesis firehose will ingest data with the least operational overhead.

upvoted 8 times

scar0909 Most Recent 1 year, 3 months ago

Selected Answer: B

Kinesis data stream cannot detined to s3
upvoted 3 times

Ruffyit 1 year, 6 months ago

1- Kinesis Data Stream provides a fully managed platform for custom data processing and analysis. Or we can say that used for custom data processing and analysis which required more manual intervention.
2- Kinesis Data Firehose simplifies the delivery of streaming data to various destinations without the need for complex transformations.
Option B is more suitable for the given scenario.
upvoted 5 times

Rhydian25 11 months, 4 weeks ago

Copy-paste from A1975's answer
upvoted 1 times

David_Ang 1 year, 7 months ago

Selected Answer: B

always if you have a service that is meant for a specific job, it the correct answer, is logic. "A" is not good enough for this situation
upvoted 3 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: B

B. Send activity data to an Amazon Kinesis Data Firehose delivery stream. Configure the stream to deliver the data to an Amazon Redshift cluster.
upvoted 3 times

NVenkatS 1 year, 9 months ago

Selected Answer: B

Petabyte scale- Redshift
upvoted 5 times

A1975 1 year, 10 months ago

Selected Answer: B

1- Kinesis Data Stream provides a fully managed platform for custom data processing and analysis. Or we can say that used for custom data processing and analysis which required more manual intervention.
2- Kinesis Data Firehose simplifies the delivery of streaming data to various destinations without the need for complex transformations.
Option B is more suitable for the given scenario.
upvoted 3 times

sickcow 1 year, 11 months ago

Selected Answer: B

Petabyte Scale sounds like Redshift!
upvoted 3 times

cookieMr 1 year, 11 months ago

Selected Answer: B

B provides a fully managed and scalable solution for data ingestion and analytics. KDF simplifies the data ingestion process by automatically scaling to handle large volumes of streaming data. It can directly load the data into an Redshift cluster, which is a powerful and fully managed data warehousing solution.

A. While Kinesis can handle streaming data, it requires additional processing to load the data into an analytics solution.

C. Although S3 and Lambda can handle the storage and processing of data, it requires more manual configuration and management compared to the fully managed solution offered by KDF and Redshift.

D. This option involves more operational overhead, as it requires managing and scaling the EC2 instances and RDS database infrastructure manually.

Therefore, option B with KDF delivering the data to Redshift cluster offers the most streamlined and operationally efficient solution for ingesting and analyzing the user activity data in the given scenario.
upvoted 3 times

pisica134 1 year, 11 months ago

petabytes in size => redshift
upvoted 3 times

mattcl 1 year, 12 months ago

It's A. Data Stream is better in this case, and you can query data in S3 with Athena
upvoted 2 times

Yadav_Sanjay 1 year, 11 months ago

Data Stream Can't write to S3. That's why B is only left correct answer.
upvoted 1 times

baba365 1 year, 11 months ago

Answer A… key phrase' least operational overhead'
KDF can write to S3 … https://docs.aws.amazon.com/firehose/latest/dev/what-is-this-service.html
upvoted 1 times

JoeGuan 1 year, 9 months ago

https://aws.amazon.com/streaming-data/ a good explanation of either option. firehose appears to be an option for Least operational overhead, as the streams product requires some building of apps etc.
upvoted 2 times

Bmarodi 2 years ago

Selected Answer: B

Option B is correct answer.
upvoted 2 times

kruasan 2 years, 1 month ago

Selected Answer: B

This solution meets the requirements as follows:
• Kinesis Data Firehose can scale to ingest and process multiple terabytes per hour of streaming data. This can easily handle the petabyte-scale data volumes.
• Firehose can deliver the data to Redshift, a petabyte-scale data warehouse, enabling on-demand SQL analytics of the data.
• Redshift is a fully managed service, minimizing operational overhead. Firehose is also fully managed, handling scalability, availability, and durability of the streaming data ingestion.
upvoted 7 times

gold4otas 2 years, 2 months ago

Selected Answer: B

B: The answer is certainly option "B" because ingesting user activity data can easily be handled by Amazon Kinesis Data streams. The ingested data can then be sent into Redshift for Analytics.

Amazon Redshift is a fully managed, petabyte-scale data warehouse service in the cloud. Amazon Redshift Serverless lets you access and analyze data without all of the configurations of a provisioned data warehouse.

https://docs.aws.amazon.com/redshift/latest/mgmt/welcome.html
upvoted 3 times

GalileoEC2 2 years, 2 months ago

the Key sentence here is: "that facilitates on-demand analytics", tthats the reason because we need to choose Kinesis Data streams over Data Firehose
upvoted 2 times

pentium75 1 year, 5 months ago

Why that? Analytics is done in Redshift, not by Kinesis

Why that? Analytics is done in Redshift, not by Kinesis.

upvoted 2 times

alexleely 2 years, 4 months ago

Selected Answer: B

B: Kinesis Data Firehose service automatically load the data into Amazon Redshift and is a petabyte-scale data warehouse service. It allows you to perform on-demand analytics with minimal operational overhead. Since the requirement didn't state what kind of analytics you need to run, we can assume that we do not need to set up additional services to provide further analytics. Thus, it has the least operational overhead.

Why not A: It is a viable solution, but storing the data in S3 would require you to set up additional services like Amazon Redshift or Amazon Athena to perform the analytics.

upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #226 *Topic 1*

A company collects data from thousands of remote devices by using a RESTful web services application that runs on an Amazon EC2 instance. The EC2 instance receives the raw data, transforms the raw data, and stores all the data in an Amazon S3 bucket. The number of remote devices will increase into the millions soon. The company needs a highly scalable solution that minimizes operational overhead.

Which combination of steps should a solutions architect take to meet these requirements? (Choose two.)

A. Use AWS Glue to process the raw data in Amazon S3. **Most Voted**

B. Use Amazon Route 53 to route traffic to different EC2 instances.

C. Add more EC2 instances to accommodate the increasing amount of incoming data.

D. Send the raw data to Amazon Simple Queue Service (Amazon SQS). Use EC2 instances to process the data.

E. Use Amazon API Gateway to send the raw data to an Amazon Kinesis data stream. Configure Amazon Kinesis Data Firehose to use the data stream as a source to deliver the data to Amazon S3. **Most Voted**

Correct Answer: *AE*

Community vote distribution

AE (100%)

Comments

Parsons Highly Voted 2 years, 4 months ago

Selected Answer: AE

A, E is the correct answer

"RESTful web services" => API Gateway.
"EC2 instance receives the raw data, transforms the raw data, and stores all the data in an Amazon S3 bucket" => GLUE with (Extract - Transform - Load)

upvoted 15 times

cookieMr Highly Voted 1 year, 11 months ago

Selected Answer: AE

A. It automatically discovers the schema of the data and generates ETL code to transform it.

E. API Gateway can be used to receive the raw data from the remote devices via RESTful web services. It provides a scalable and managed infrastructure to handle the incoming requests. The data can then be sent to an Amazon Kinesis data stream, which is a highly scalable and durable real-time data streaming service. From there, Amazon Kinesis Data Firehose can be configured to use the data stream as a source and deliver the transformed data to Amazon S3. This combination of services allows for the seamless ingestion and processing of data while minimizing operational overhead.

B. It does not directly address the need for scalable data processing and storage. It focuses on managing DNS and routing traffic to different endpoints.

C. Adding more EC2 can lead to increased operational overhead in terms of managing and scaling the instances.

D. Using SQS and EC2 for processing data introduces more complexity and operational overhead.
upvoted 5 times

Noveo Most Recent 8 months, 4 weeks ago

AE breaks the original workflow "receive raw data - process - store" to "receive - store- process - store again" which leads to additional storage consuming (and thus money consuming).
upvoted 2 times

Ruffyit 1 year, 6 months ago

A - Use AWS Glue to process the raw data in Amazon S3
E - Use Amazon API Gateway to send the raw data to an Amazon Kinesis data stream. Configure Amazon Kinesis Data Firehose to use the data stream as a source to deliver the data to Amazon S3
upvoted 2 times

TariqKipkemei 1 year, 8 months ago

Selected Answer: AE

E then A no doubt.
upvoted 2 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: AE

A. It automatically discovers the schema of the data and generates ETL code to transform it.

E. API Gateway can be used to receive the raw data from the remote devices via RESTful web services. It provides a scalable and managed infrastructure to handle the incoming requests. The data can then be sent to an Amazon Kinesis data stream, which is a highly scalable and durable real-time data streaming service. From there, Amazon Kinesis Data Firehose can be configured to use the data stream as a source and deliver the transformed data to Amazon S3. This combination of services allows for the seamless ingestion and processing of data while minimizing operational overhead.
upvoted 2 times

ibu007 1 year, 9 months ago

Selected Answer: AE

A - Use AWS Glue to process the raw data in Amazon S3
E - Use Amazon API Gateway to send the raw data to an Amazon Kinesis data stream. Configure Amazon Kinesis Data Firehose to use the data stream as a source to deliver the data to Amazon S3
upvoted 3 times

GCB1990 1 year, 9 months ago

Correct answer: D and E
upvoted 1 times

wRhlH 1 year, 11 months ago

Why not BC?
upvoted 1 times

AnnieTran_91 1 year, 12 months ago

Why it not CE?
Add more EC2 instances to accommodate the increasing amount of incoming data?
upvoted 1 times

TTaws 1 year, 11 months ago

EC2 is not server-less. they want to minimize overhead
upvoted 1 times

studynoplay 2 years ago

Selected Answer: AE

minimizes operational overhead = Serverless
Glue, Kinesis Datastream, S3 are serverless
upvoted 3 times

1e22522 10 months, 1 week ago

facts!!
upvoted 1 times

KZM 2 years, 3 months ago

How about "C" to increase EC2 instances for the increased devices soon?
upvoted 1 times

Aninina 2 years, 4 months ago

Selected Answer: AE

Glue and API
upvoted 3 times

mhmt4438 2 years, 4 months ago

Selected Answer: AE

https://www.examtopics.com/discussions/amazon/view/83387-exam-aws-certified-solutions-architect-associate-saa-c02/
upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #227 *Topic 1*

A company needs to retain its AWS CloudTrail logs for 3 years. The company is enforcing CloudTrail across a set of AWS accounts by using AWS Organizations from the parent account. The CloudTrail target S3 bucket is configured with S3 Versioning enabled. An S3 Lifecycle policy is in place to delete current objects after 3 years.

After the fourth year of use of the S3 bucket, the S3 bucket metrics show that the number of objects has continued to rise. However, the number of new CloudTrail logs that are delivered to the S3 bucket has remained consistent.

Which solution will delete objects that are older than 3 years in the MOST cost-effective manner?

A. Configure the organization's centralized CloudTrail trail to expire objects after 3 years.

B. Configure the S3 Lifecycle policy to delete previous versions as well as current versions. **Most Voted**

C. Create an AWS Lambda function to enumerate and delete objects from Amazon S3 that are older than 3 years.

D. Configure the parent account as the owner of all objects that are delivered to the S3 bucket.

Correct Answer: *B*

Community vote distribution

B (94%) C (6%)

Comments

Guru4Cloud Highly Voted 1 year, 2 months ago

Selected Answer: B

This is the most cost-effective option because:
• Versioning has caused the number of objects to increase over time, even as current objects are deleted after 3 years. By deleting previous versions as well, this will clean up old object versions and reduce storage costs.
• An S3 Lifecycle policy incurs no additional charges and requires no additional resources to configure and run. It is a native S3 tool for managing object lifecycles cost-effectively.

upvoted 11 times

cookieMr Highly Voted 1 year, 5 months ago

Selected Answer: B

By configuring the S3 Lifecycle policy to delete previous versions as well as current versions, the older versions of the CloudTrail logs will be deleted. This ensures that objects older than 3 years are removed from the S3 bucket, reducing the object count and controlling storage costs.

and controlling storage costs.

A. This option is not directly related to managing objects in the S3. It focuses on configuring the expiration of CloudTrail trails, which may not address the need to delete objects from the S3 bucket.

C. While it is technically possible to create a Lambda to delete objects older than 3 years, this approach would introduce additional complexity and operational overhead.

D. Changing the ownership of the objects in the S3 bucket does not directly address the need to delete objects older than 3 years. Ownership does not affect the deletion behavior of the objects.

upvoted 5 times

Mikado211 Most Recent 1 year ago

Selected Answer: B

I did something similar recently : Lifecycle is triggered more or less each 24 hours, in my case it removed hundreds of gigabytes and millions of small files in one shot. Using another mechanism like a script would have taken days if not weeks.

upvoted 3 times

Ruffyit 1 year ago

This is the most cost-effective option because:
• Versioning has caused the number of objects to increase over time, even as current objects are deleted after 3 years. By deleting previous versions as well, this will clean up old object versions and reduce storage costs.
• An S3 Lifecycle policy incurs no additional charges and requires no additional resources to configure and run. It is a native S3 tool for managing object lifecycles cost-effectively.

upvoted 3 times

TariqKipkemei 1 year, 2 months ago

Selected Answer: B

Ensure to delete previous versions as well.

upvoted 2 times

Bmarodi 1 year, 6 months ago

Selected Answer: B

I go for option B.

upvoted 2 times

ruqui 1 year, 6 months ago

I don't think it's possible to configure an S3 lifecycle policy to delete all versions of an object, so B is wrong ... I think the question is improperly worded

upvoted 2 times

Rahulbit34 1 year, 7 months ago

• Versioning has caused the number of objects to increase over time, even as current objects are deleted after 3 years. By deleting previous versions as well, this will clean up old object versions and reduce storage costs. • An S3 Lifecycle policy incurs no additional charges and requires no additional resources to configure and run. It is a native S3 tool for managing object lifecycles cost-effectively.

upvoted 2 times

kruasan 1 year, 7 months ago

Selected Answer: B

This is the most cost-effective option because:
• Versioning has caused the number of objects to increase over time, even as current objects are deleted after 3 years. By deleting previous versions as well, this will clean up old object versions and reduce storage costs.
• An S3 Lifecycle policy incurs no additional charges and requires no additional resources to configure and run. It is a native S3 tool for managing object lifecycles cost-effectively.

upvoted 4 times

kruasan 1 year, 7 months ago

https://docs.aws.amazon.com/AmazonS3/latest/userguide/DeletingObjectVersions.html

upvoted 3 times

bullrem 1 year, 10 months ago

Selected Answer: C

A more cost-effective solution would be to configure the organization's centralized CloudTrail trail to expire objects after 3 years. This would ensure that all objects, including previous versions, are deleted after the specified retention period.
Another option would be to create an AWS Lambda function to enumerate and delete objects from Amazon S3 that are older than 3 years, this would allow you to have more control over the deletion process and to write a custom logic that best fits your use case.

upvoted 3 times

pentium75 11 months, 2 weeks ago

As long as versioning on the S3 bucket is enabled, any deletion, whether performed by CloudTrail or by your custom Lambda function, will simply add a new version with a deletion market but will not delete the previous version.

upvoted 3 times

JayBee65 1 year, 10 months ago

Selected Answer: B

The question clearly says "An S3 Lifecycle policy is in place to delete current objects after 3 years". This implies that previous versions are not deleted, since this is a separate setting, and since logs are constantly changed, it would seem to make sense to delete previous versions so, so B. D is wrong, since the parent account (the management account) will already be the owner of all objects delivered to the S3 bucket, "All accounts in the organization can see MyOrganizationTrail in their list of trails, but member accounts cannot remove or modify the organization trail. Only the management account or delegated administrator account can change or delete the trail for the organization.", see
https://docs.aws.amazon.com/awscloudtrail/latest/userguide/creating-trail-organization.html

upvoted 3 times

John_Zhuang 1 year, 10 months ago

Selected Answer: B

B is the right answer. Ref: https://docs.aws.amazon.com/awscloudtrail/latest/userguide/best-practices-security.html#:~:text=The%20CloudTrail%20trail,time%20has%20passed.

Option A is wrong. No way to expire the cloudtrail logs

upvoted 4 times

techhb 1 year, 10 months ago

Selected Answer: B

Configure the S3 Lifecycle policy to delete previous versions

upvoted 3 times

Aninina 1 year, 10 months ago

Selected Answer: B

B. Configure the S3 Lifecycle policy to delete previous versions as well as current versions.

upvoted 2 times

Aninina 1 year, 10 months ago

B. Configure the S3 Lifecycle policy to delete previous versions as well as current versions.

upvoted 2 times

Parsons 1 year, 10 months ago

Selected Answer: B

B is correct answer

upvoted 3 times

AHUI 1 year, 10 months ago

Ans: A
https://docs.aws.amazon.com/awscloudtrail/latest/userguide/creating-trail-organization.html
When you create an organization trail, a trail with the name that you give it is created in every AWS account that belongs to your organization. Users with CloudTrail permissions in member accounts can see this trail when they log into the AWS CloudTrail console from their AWS accounts, or when they run AWS CLI commands such as describe-trail. However, users in member accounts do not have sufficient permissions to delete the organization trail, turn logging on or off, change what types of events are logged, or otherwise change the organization trail in any way.

upvoted 1 times

upvoted 1 times

AHUI 1 year, 10 months ago

correction: Ans D is the answer.
https://docs.aws.amazon.com/awscloudtrail/latest/userguide/creating-trail-organization.html
upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #228 *Topic 1*

A company has an API that receives real-time data from a fleet of monitoring devices. The API stores this data in an Amazon RDS DB instance for later analysis. The amount of data that the monitoring devices send to the API fluctuates. During periods of heavy traffic, the API often returns timeout errors.

After an inspection of the logs, the company determines that the database is not capable of processing the volume of write traffic that comes from the API. A solutions architect must minimize the number of connections to the database and must ensure that data is not lost during periods of heavy traffic.

Which solution will meet these requirements?

A. Increase the size of the DB instance to an instance type that has more available memory.

B. Modify the DB instance to be a Multi-AZ DB instance. Configure the application to write to all active RDS DB instances.

C. Modify the API to write incoming data to an Amazon Simple Queue Service (Amazon SQS) queue. Use an AWS Lambda function that Amazon SQS invokes to write data from the queue to the database. **Most Voted**

D. Modify the API to write incoming data to an Amazon Simple Notification Service (Amazon SNS) topic. Use an AWS Lambda function that Amazon SNS invokes to write data from the topic to the database.

Correct Answer: *C*

Community vote distribution

C (100%)

Comments

toyaji Highly Voted 9 months, 2 weeks ago

Selected Answer: C

You need to also use AWS RDS Proxy becuase lambda will increase parallel and it will cause connection error

upvoted 5 times

mwwt2022 Most Recent 1 year, 5 months ago

Selected Answer: C

//minimize the number of connections to the database and must ensure that data is not lost during periods of heavy traffic//

I go for C
upvoted 3 times

Ruffyit 1 year, 6 months ago

Decouple the API and the DB with Amazon Simple Queue Service (Amazon SQS) queue.
upvoted 3 times

oluolope 1 year, 7 months ago

https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/sqs-configure-lambda-function-trigger.html
SQS can invoke lambda indeed. Initially I picked D because I wasn't sure it was possible but , this article shows it is. It makes this question even more confusing for me as it is also possible to trigger lambda from SNS:
https://docs.aws.amazon.com/sns/latest/dg/sns-lambda-as-subscriber.html
I don't know which option between C and D makes more sense. I still have a preference for D as it seems less hacky than C.
upvoted 1 times

TariqKipkemei 1 year, 8 months ago

Selected Answer: C

Decouple the API and the DB with Amazon Simple Queue Service (Amazon SQS) queue.
upvoted 4 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: C

C. Modify the API to write incoming data to an Amazon Simple Queue Service (Amazon SQS) queue. Use an AWS Lambda function that Amazon SQS invokes to write data from the queue to the database.
upvoted 2 times

cookieMr 1 year, 11 months ago

Selected Answer: C

By leveraging SQS as a buffer and using an Lambda to process and write data from the queue to the database, the solution provides scalability, decoupling, and reliability while minimizing the number of connections to the database. This approach handles fluctuations in traffic and ensures data integrity during high-traffic periods.

A. Increasing the size of the DB instance may provide more memory, but it does not address the issue of handling high write traffic efficiently and minimizing connections to the database.

B. Modifying the DB instance to be a Multi-AZ instance and writing to all active instances can improve availability but does not address the issue of efficiently handling high write traffic and minimizing connections to the database.

D. Using SNS and an Lambda can provide decoupling and scalability, but it is not suitable for handling heavy write traffic efficiently and minimizing connections to the database.
upvoted 3 times

Moccorso 1 year, 11 months ago

I think D, "Use an AWS Lambda function that Amazon SQS invokes to write data from the queue to the database" SQS can't invokes Lambda becouse SQS is pull.
upvoted 3 times

pentium75 1 year, 5 months ago

"Invoke a Lambda function from an Amazon SQS trigger"
https://docs.aws.amazon.com/lambda/latest/dg/example_serverless_SQS_Lambda_section.html
upvoted 2 times

shivamrulz 1 year, 11 months ago

Why not B
upvoted 2 times

pentium75 1 year, 5 months ago

a) You can't write to multiple instances at the same time
b) If you could, it would probably not increase performance
c) if it would increase performance then it would probably double performance, which might not be enough

c) If it would increase performance then it would probably double performance, which might not be enough.

Decoupling is the way to go - let clients submit data via APIs, and write it asynchronously to the database. Don't let clients wait until data has been written.
upvoted 2 times

Russs99 2 years, 2 months ago

C is in deed the correct answer for the use case
upvoted 2 times

kaushald 2 years, 3 months ago

Selected Answer: C

C is correct
upvoted 2 times

Steve_4542636 2 years, 3 months ago

Selected Answer: C

Cis correct
upvoted 2 times

maciekmaciek 2 years, 3 months ago

Selected Answer: C

C looks ok
upvoted 2 times

iamjaehyuk 2 years, 4 months ago

why not D?
upvoted 1 times

Parsons 2 years, 4 months ago

Selected Answer: C

C is correct.
upvoted 2 times

mhmt4438 2 years, 4 months ago

Selected Answer: C

C. Modify the API to write incoming data to an Amazon Simple Queue Service (Amazon SQS) queue. Use an AWS Lambda function that Amazon SQS invokes to write data from the queue to the database.

To minimize the number of connections to the database and ensure that data is not lost during periods of heavy traffic, the company should modify the API to write incoming data to an Amazon SQS queue. The use of a queue will act as a buffer between the API and the database, reducing the number of connections to the database. And the use of an AWS Lambda function invoked by SQS will provide a more flexible way of handling the data and processing it. This way, the function will process the data from the queue and insert it into the database in a more controlled way.
upvoted 3 times

Aninina 2 years, 4 months ago

Did you use ChatGPT?
upvoted 6 times

Nguyen25183 2 years, 3 months ago

same question as you :D
upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #229 *Topic 1*

A company manages its own Amazon EC2 instances that run MySQL databases. The company is manually managing replication and scaling as demand increases or decreases. The company needs a new solution that simplifies the process of adding or removing compute capacity to or from its database tier as needed. The solution also must offer improved performance, scaling, and durability with minimal effort from operations.

Which solution meets these requirements?

A. Migrate the databases to Amazon Aurora Serverless for Aurora MySQL. **Most Voted**

B. Migrate the databases to Amazon Aurora Serverless for Aurora PostgreSQL.

C. Combine the databases into one larger MySQL database. Run the larger database on larger EC2 instances.

D. Create an EC2 Auto Scaling group for the database tier. Migrate the existing databases to the new environment.

Correct Answer: *A*

Community vote distribution

A (100%)

Comments

cookieMr Highly Voted 11 months, 2 weeks ago

Selected Answer: A

Migrating the databases to Aurora Serverless provides automated scaling and replication capabilities. Aurora Serverless automatically scales the capacity based on the workload, allowing for seamless addition or removal of compute capacity as needed. It also offers improved performance, durability, and high availability without requiring manual management of replication and scaling.

B. Incorrect because it suggests migrating to a different database engine, which may introduce compatibility issues and require significant code modifications.

C. Incorrect because consolidating into a larger MySQL database on larger EC2 instances does not provide the desired scalability and automation.

D. Incorrect because using EC2 Auto Scaling groups for the database tier still requires manual management of replication and scaling.

upvoted 9 times

satyaammm Most Recent 3 months, 2 weeks ago

Selected Answer: A

Aurora Serverless is the most suitable option here.
upvoted 1 times

TariqKipkemei 8 months, 2 weeks ago

Selected Answer: A

Migrate the databases to Amazon Aurora Serverless for Aurora MySQL
upvoted 2 times

Undisputed 10 months, 2 weeks ago

Selected Answer: A

Aurora MySQL
upvoted 2 times

Bmarodi 1 year ago

Selected Answer: A

Option A is right answer.
upvoted 2 times

Bhrino 1 year, 3 months ago

Selected Answer: A

A is correct because aurora might be more expensive but its serverless and is much faster
upvoted 2 times

mp165 1 year, 4 months ago

Selected Answer: A

A is porper

https://aws.amazon.com/rds/aurora/serverless/
upvoted 4 times

Aninina 1 year, 4 months ago

Selected Answer: A

Aurora MySQL
upvoted 2 times

mhmt4438 1 year, 4 months ago

Selected Answer: A

https://www.examtopics.com/discussions/amazon/view/51509-exam-aws-certified-solutions-architect-associate-saa-c02/
upvoted 2 times

EXAMTOPICS

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #230 *Topic 1*

A company is concerned that two NAT instances in use will no longer be able to support the traffic needed for the company's application. A solutions architect wants to implement a solution that is highly available, fault tolerant, and automatically scalable.

What should the solutions architect recommend?

A. Remove the two NAT instances and replace them with two NAT gateways in the same Availability Zone.

B. Use Auto Scaling groups with Network Load Balancers for the NAT instances in different Availability Zones.

C. Remove the two NAT instances and replace them with two NAT gateways in different Availability Zones. **Most Voted**

D. Replace the two NAT instances with Spot Instances in different Availability Zones and deploy a Network Load Balancer.

Correct Answer: *C*

Community vote distribution

C (97%) B

Comments

Bhrino Highly Voted 1 year, 9 months ago

Selected Answer: C

fyi yall in most cases nat instances are a bad thing because their customer managed while nat gateways are AWS Managed. So in this case I already know to get rid of the nat instances the reason its c is because it wants high availability meaning different AZs

upvoted 7 times

pentium75 Highly Voted 11 months, 2 weeks ago

Selected Answer: C

See https://docs.aws.amazon.com/vpc/latest/userguide/vpc-nat-comparison.html

upvoted 7 times

bora4motion Most Recent 1 month ago

Selected Answer: C

c is correct, nat instances are obsolete anyways

upvoted 1 times

satyaammm 3 months, 2 weeks ago

Selected Answer: C

NAT Gateways in 2 different AZ's is the most suitable option here.

upvoted 1 times

MiniYang 1 year ago

Selected Answer: B

Highly available, fault tolerant and automatically scalable=> Autoscaling and Diffrent AZ

upvoted 1 times

pentium75 11 months, 2 weeks ago

NAT instances are legacy technology. "If you're already using a NAT instance, we recommend that you replace it with a NAT gateway." Thus C.

https://docs.aws.amazon.com/vpc/latest/userguide/vpc-nat-comparison.html

upvoted 3 times

TariqKipkemei 1 year, 2 months ago

Selected Answer: C

Highly available, fault tolerant, and automatically scalable = two NAT gateways in different Availability Zones

upvoted 3 times

Undisputed 1 year, 4 months ago

Selected Answer: C

Remove the two NAT instances and replace them with two NAT gateways in different Availability Zones

upvoted 2 times

cookieMr 1 year, 5 months ago

Selected Answer: C

This recommendation ensures high availability and fault tolerance by distributing the NAT gateways across multiple AZs. NAT gateways are managed AWS services that provide scalable and highly available outbound NAT functionality. By deploying NAT gateways in differentAZs, the company can achieve redundancy and avoid a single point of failure. This solution also provides automatic scaling to handle increasing traffic without manual intervention.

Option A is incorrect because placing both NAT gateways in the same Availability Zone does not provide fault tolerance.

Option B is incorrect because using Auto Scaling groups with Network Load Balancers is not the recommended approach for NAT instances.

Option D is incorrect because Spot Instances are not suitable for critical infrastructure components like NAT instances.

upvoted 5 times

Axeashes 1 year, 6 months ago

Selected Answer: C

HA: https://docs.aws.amazon.com/vpc/latest/userguide/vpc-nat-comparison.html
Scalability: https://docs.aws.amazon.com/vpc/latest/userguide/vpc-nat-gateway.html

upvoted 2 times

Theodorz 1 year, 10 months ago

Could anybody teach me why the B cannot be correct answer? This solution also seems providing Scalability(Auto Scaling Group), High Availability(different AZ), and Fault Tolerance(NLB & AZ).

I honestly think that C is not enough, because each NAT gateway can provide a few scalability, but the bandwidth limit is clearly explained in the document. The C exactly mentioned "two NAT gateways" so the number of NAT is fixed, which will reach its limit soon.

upvoted 3 times

KZM 1 year, 9 months ago

Option B proposes to use an Auto Scaling group with Network Load Balancers to continue using the existing two NAT

Option B proposes to use an Auto Scaling group with Network Load Balancers to continue using the existing two NAT instances. However, NAT instances do not support automatic failover without a script, unlike NAT gateways which provide this functionality. Additionally, using Network Load Balancers to balance traffic between NAT instances adds more complexity to the solution.

https://docs.aws.amazon.com/vpc/latest/userguide/vpc-nat-comparison.html

upvoted 3 times

mwwt2022 11 months, 2 weeks ago

Thx for your explanation!

upvoted 1 times

JayBee65 1 year, 10 months ago

C. If you have resources in multiple Availability Zones and they share one NAT gateway, and if the NAT gateway's Availability Zone is down, resources in the other Availability Zones lose internet access. To create an Availability Zone-independent architecture, create a NAT gateway in each Availability Zone and configure your routing to ensure that resources use the NAT gateway in the same Availability Zone. https://docs.aws.amazon.com/vpc/latest/userguide/vpc-nat-gateway.html#nat-gateway-basics

upvoted 3 times

techhb 1 year, 10 months ago

Selected Answer: C

Replace NAT Instances with Gateway

upvoted 2 times

mhmt4438 1 year, 10 months ago

Selected Answer: C

Correct answer is C

upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #231 *Topic 1*

An application runs on an Amazon EC2 instance that has an Elastic IP address in VPC A. The application requires access to a database in VPC B. Both VPCs are in the same AWS account.

Which solution will provide the required access MOST securely?

A. Create a DB instance security group that allows all traffic from the public IP address of the application server in VPC A.

B. Configure a VPC peering connection between VPC A and VPC B. **Most Voted**

C. Make the DB instance publicly accessible. Assign a public IP address to the DB instance.

D. Launch an EC2 instance with an Elastic IP address into VPC B. Proxy all requests through the new EC2 instance.

Correct Answer: *B*

Community vote distribution

B (91%) A (9%)

Comments

JayBee65 Highly Voted 2 years, 4 months ago

A is correct. B will work but is not the most secure method, since it will allow everything in VPC A to talk to everything in VPC B and vice versa, not at all secure. A on the other hand will only allow the application (since you select it's IP address) to talk to the application server in VPC A - you are allowing only the required connectivity. See the link for this exact use case: https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/Overview.RDSSecurityGroups.html

upvoted 14 times

mhmt4438 2 years, 4 months ago

" allows all traffic from the public IP address" Nice bro niceee This is absolutely the most secure method at all. :)))

upvoted 17 times

test_devops_aws 2 years, 2 months ago

:))))))))

upvoted 1 times

datz 2 years, 2 months ago

he must be the security engineer lolol :D

"Jaybee" - Please dont ever say that traffic over the public internet is secure :D
upvoted 5 times

swakan 6 months, 2 weeks ago

https://aws.amazon.com/vpc/faqs/
upvoted 1 times

graveend 1 year, 10 months ago

Both VPCs are in the "SAME AWS ACCOUNT" and the requirement specifies allowing traffic from the *PUBLIC IP of the APPLICATION SERVER*. In this case the traffic remains inside the AWS infrastructure or will it go through the public internet?
upvoted 2 times

pentium75 1 year, 5 months ago

Answer A (not "the requirement") specifies "allowing traffic from the public IP", which is for sure NOT the "most secure" option.
upvoted 2 times

DUBURA Highly Voted 1 year, 6 months ago

Selected Answer: B

B. Configure a VPC peering connection between VPC A and VPC B.

The most secure solution is to configure a VPC peering connection between the two VPCs. This allows private communication between the application server and the database, without exposing resources to the public internet.

Option A exposes the database to the public internet by allowing inbound traffic from a public IP address.

Option C makes the database instance itself public, which is insecure.

Option D adds complexity with a proxy that is not needed when a VPC peering connection can enable private communication between VPCs.

So option B is the most secure while allowing the necessary connectivity between the application server and the database in the separate VPCs.
upvoted 12 times

FlyingHawk Most Recent 4 months, 2 weeks ago

Selected Answer: B

It should be B, plus the DB security g A. group that allows all traffic from the security group of the application server in VPC A
upvoted 1 times

Jazz888 1 year ago

Well this is a tricky one!!! Are we going to assume that the database in VPC B is in private subnet? In that case configuring security group to allow the traffic coming from Elastic IP of VPC A will not work. And if we use peering, the resources that live in the same subnet as the EC2 instance in VPC A will have access to the database? So what would we say to this? Is moving traffic through the public AWS space is safer than allowing access to the DB to other resources in VPC A?..... I don't know what to think
upvoted 2 times

Ruffyit 1 year, 6 months ago

When you establish peering relationships between VPCs across different AWS Regions, resources in the VPCs (for example, EC2 instances and Lambda functions) in different AWS Regions can communicate with each other using private IP addresses, without using a gateway, VPN connection, or network appliance. The traffic remains in the private IP space. All inter-Region traffic is encrypted with no single point of failure, or bandwidth bottleneck. Traffic always stays on the global AWS backbone, and never traverses the public internet, which reduces threats, such as common exploits, and DDoS attacks. Inter-Region VPC peering provides a simple and cost-effective way to share resources between regions or replicate data for geographic redundancy.
upvoted 2 times

rlamberti 1 year, 7 months ago

Selected Answer: B

Most secure = not leaving AWS network.
VPC peering is the way.

VPC peering is the way.

upvoted 3 times

TariqKipkemei 1 year, 8 months ago

Selected Answer: B

VPC to VPC comms = VPC peering

upvoted 2 times

Sutariya 1 year, 9 months ago

B is correct : Setup VPC peering and connect Application from VPC A to connect with VPC B in private subnet so DB instace always secure with internet.

upvoted 2 times

_d1rk_ 1 year, 9 months ago

Am I missing something or simply A is wrong because, without VPC peering (or other inter-connection sharing mechanisms such as Transit Gateway or VPN), VPC A and VPC B cannot communicate each other?

upvoted 1 times

jacob_ho 1 year, 9 months ago

can use vpc endpoints but no option use that

upvoted 1 times

A1975 1 year, 10 months ago

Selected Answer: B

When you establish peering relationships between VPCs across different AWS Regions, resources in the VPCs (for example, EC2 instances and Lambda functions) in different AWS Regions can communicate with each other using private IP addresses, without using a gateway, VPN connection, or network appliance. The traffic remains in the private IP space. All inter-Region traffic is encrypted with no single point of failure, or bandwidth bottleneck. Traffic always stays on the global AWS backbone, and never traverses the public internet, which reduces threats, such as common exploits, and DDoS attacks. Inter-Region VPC peering provides a simple and cost-effective way to share resources between regions or replicate data for geographic redundancy.

https://docs.aws.amazon.com/vpc/latest/peering/what-is-vpc-peering.html

upvoted 4 times

animefan1 1 year, 11 months ago

Selected Answer: B

With peering, we EC2 can communicate with RDS. RDS SG can have inbound from EC2 IP rather than VPC CIDR for more security

upvoted 2 times

maggie135 1 year, 11 months ago

Selected Answer: B

VPC peering uses AWS network.

upvoted 2 times

cookieMr 1 year, 11 months ago

Selected Answer: B

By configuring a VPC peering connection between VPC A and VPC B, you can establish private and secure communication between the EC2 instance in VPC A and the database in VPC B. VPC peering allows traffic to flow between the two VPCs using private IP addresses, without the need for public IP addresses or exposing the database to the internet.

Option A is not the best solution as it requires allowing all traffic from the public IP address of the application server, which can be less secure.

Option C involves making the DB instance publicly accessible, which introduces security risks by exposing the database directly to the internet.

Option D adds unnecessary complexity by launching an additional EC2 instance in VPC B and proxying all requests through it, which is not the most efficient and secure approach in this scenario.

upvoted 5 times

joechen2023 1 year, 11 months ago

Selected Answer: B

I don't like A because the security group setting is wrong as it set up to allow all public IP addresses. If the security group setting is correct, then I will go for A
I don't like B because it need to set up security group as well on top of peering.
for exam purpose only, I will go with the least worst choice which is B

upvoted 1 times

Bmarodi 1 year, 12 months ago

Selected Answer: A

The keywords are: "access MOST securely", hence the option A meets these requirements.

upvoted 1 times

smartegnine 1 year, 12 months ago

Selected Answer: A

Each VPC security group rule makes it possible for a specific source to access a DB instance in a VPC that is associated with that VPC security group. The source can be a range of addresses (for example, 203.0.113.0/24), or another VPC security group.

https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide

upvoted 1 times

MostafaWardany 2 years ago

Selected Answer: B

Most secure = VPC peering

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #232 *Topic 1*

A company runs demonstration environments for its customers on Amazon EC2 instances. Each environment is isolated in its own VPC. The company's operations team needs to be notified when RDP or SSH access to an environment has been established.

A. Configure Amazon CloudWatch Application Insights to create AWS Systems Manager OpsItems when RDP or SSH access is detected.

B. Configure the EC2 instances with an IAM instance profile that has an IAM role with the AmazonSSMManagedInstanceCore policy attached.

C. Publish VPC flow logs to Amazon CloudWatch Logs. Create required metric filters. Create an Amazon CloudWatch metric alarm with a notification action for when the alarm is in the ALARM state. **Most Voted**

D. Configure an Amazon EventBridge rule to listen for events of type EC2 Instance State-change Notification. Configure an Amazon Simple Notification Service (Amazon SNS) topic as a target. Subscribe the operations team to the topic.

Correct Answer: *C*

Community vote distribution

C (84%) Other (16%)

Comments

cookieMr Highly Voted 1 year, 5 months ago

Selected Answer: C

By publishing VPC flow logs to CloudWatch Logs and creating metric filters to detect RDP or SSH access, the operations team can configure an CloudWatch metric alarm to notify them when the alarm is triggered. This will provide the desired notification when RDP or SSH access to an environment is established.

Option A is incorrect because CloudWatch Application Insights is not designed for detecting RDP or SSH access.

Option B is also incorrect because configuring an IAM instance profile with the AmazonSSMManagedInstanceCore policy does not directly address the requirement of notifying the operations team when RDP or SSH access occurs.

Option D is wrong beacuse configuring an EventBridge rule to listen for EC2 Instance State-change Notification events and using an SNS topic as a target will notify the operations team about changes in the instance state, such as starting or stopping instances. However, it does not specifically detect or notify when RDP or SSH access is established, which is the requirement stated in the question.

upvoted 16 times

Vickysss Highly Voted 1 year, 10 months ago

Selected Answer: C

https://aws.amazon.com/blogs/security/how-to-monitor-and-visualize-failed-ssh-access-attempts-to-amazon-ec2-linux-instances/

upvoted 9 times

NitiATOS 1 year, 10 months ago

https://docs.aws.amazon.com/vpc/latest/userguide/flow-logs-records-examples.html#flow-log-example-accepted-rejected

Adding this to support that VPC flow logs can be used to cvapture Accepted or Rejected SSH and RDP traffic.

upvoted 4 times

ruqui 1 year, 6 months ago

I don't think C would be an acceptable solution ... the request is to be notified WHEN a SSH and/or RDP connection is established so it requires real-time monitoring and that is something the C solution does not provide ... I would select A as a correct answer

upvoted 1 times

0xE8D4A51000 Most Recent 6 months ago

Selected Answer: D

See https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/monitoring-instance-state-changes.html

upvoted 1 times

pentium75 11 months ago

Selected Answer: C

C sounds complex, but is the only answer that can work.
Not A - Application Insights has nothing to do with SSH/RDP access to the OS; also we need a notification, not an OpsItem
Not B - Just attaching a role does not create a notification
Not D - Establishing SSH/RDP access is not a "state change" that would trigger this

upvoted 3 times

pentium75 11 months, 2 weeks ago

Selected Answer: C

A bit clueless here. AWS-recommended approach involves the CloudWatch Logs Agent on each EC2 instance, but that is not involved in any of the answers.

A: Sounds good at first read, but "CloudWatch Application Insights" cannot detect RDP or SSH access.

B: Would allow RDP or SSH access via Systems Manager, but would NOT prevent access without Systems Manager; also we'd need to configure notifications in Systems Manager which is not mentioned here.

upvoted 3 times

pentium75 11 months, 2 weeks ago

C: Could work but it seems overkill to capture VPC flow logs just to detect SSH and RDP traffic. Also it is not real-time, and it's unclear how and when exactly the state transitions and notifications will be triggered. At best you'd get notification few minutes AFTER (not "when") "access has been established". Still, is has most similarity with the recommended approach to detect failed connections: https://aws.amazon.com/tr/blogs/security/how-to-monitor-and-visualize-failed-ssh-access-attempts-to-amazon-ec2-linux-instances/

D: Won't work because establishment of a connection is not an instance state change.

upvoted 3 times

Ruffyit 1 year ago

https://docs.aws.amazon.com/vpc/latest/userguide/flow-logs-records-examples.html#flow-log-example-accepted-rejected

Adding this to support that VPC flow logs can be used to cvapture Accepted or Rejected SSH and RDP traffic.

upvoted 2 times

TariqKipkemei 1 year, 2 months ago

Selected Answer: C

Publish VPC flow logs to Amazon CloudWatch Logs. Create required metric filters. Create an Amazon CloudWatch metric alarm with a notification action for when the alarm is in the ALARM state

upvoted 2 times

Bmarodi 1 year, 5 months ago

Selected Answer: C

VPC Flow Logs is a feature that enables you to capture information about the IP traffic going to and from network interfaces in your VPC. Flow log data can be published to the following locations: Amazon CloudWatch Logs, Amazon S3, or Amazon Kinesis Data Firehose. After you create a flow log, you can retrieve and view the flow log records in the log group, bucket, or delivery stream that you configured.

Flow logs can help you with a number of tasks, such as:

Diagnosing overly restrictive security group rules

Monitoring the traffic that is reaching your instance

Determining the direction of the traffic to and from the network interfaces
Ref link: https://docs.aws.amazon.com/vpc/latest/userguide/flow-logs.html

upvoted 4 times

cokutan 1 year, 6 months ago

Selected Answer: C

seems like c:
https://aws.amazon.com/tr/blogs/security/how-to-monitor-and-visualize-failed-ssh-access-attempts-to-amazon-ec2-linux-instances/

upvoted 2 times

pentium75 11 months, 2 weeks ago

This link does not mention VPC flow logs at all.

upvoted 1 times

ChrisAn 1 year, 6 months ago

Selected Answer: D

D. Configure an Amazon EventBridge rule to listen for events of type EC2 Instance State-change Notification. Configure an Amazon Simple Notification Service (Amazon SNS) topic as a target. Subscribe the operations team to the topic. This setup allows the EventBridge rule to capture instance state change events, such as when RDP or SSH access is established. The rule can then send notifications to the specified SNS topic, which is subscribed by the operations team.

upvoted 2 times

markw92 1 year, 5 months ago

D is wrong. EC2 instance state change is only for pending, running etc. https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/monitoring-instance-state-changes.html you can't have state change of ssh or rdp.

upvoted 1 times

datz 1 year, 8 months ago

Selected Answer: C

C:

https://www.youtube.com/watch?v=KAe3Eju59OU

upvoted 2 times

Abhineet9148232 1 year, 9 months ago

Selected Answer: C

https://aws.amazon.com/blogs/security/how-to-monitor-and-visualize-failed-ssh-access-attempts-to-amazon-ec2-linux-instances/

upvoted 2 times

bullrem 1 year, 10 months ago

Selected Answer: A

A. Configuring Amazon CloudWatch Application Insights to create AWS Systems Manager OpsItems when RDP or SSH access is detected would be the most appropriate solution in this scenario. This would allow the operations team to be notified when RDP or SSH access has been established and provide them with the necessary information to take action if needed. Additionally, Amazon CloudWatch Application Insights would allow for monitoring and troubleshooting of the system in real-time.

upvoted 1 times

Training4aBetterLife 1 year, 10 months ago

Selected Answer: C

EC2 Instance State-change Notifications are not the same as RDP or SSH established connection notifications. Use Amazon CloudWatch Logs to monitor SSH access to your Amazon EC2 Linux instances so that you can monitor rejected (or established) SSH connection requests and take action.

upvoted 5 times

alexleely 1 year, 10 months ago

Selected Answer: A

The Answer can be A or C depending on the requirement if it requires real-time notification.
A: Allows the operations team to be notified in real-time when access is established, and also provides visibility into the access events through the OpsItems.

C: The logs will need to be analyzed and metric filters applied to detect access, and then the alarm will trigger based on that analysis. This method could have a delay in providing notifications. Thus, not the best solution if real-time notification is required.

Why not D: RDP or SSH access does not cause an EC2 instance to have a state change. The state change events that Amazon EventBridge can listen for include stopping, starting, and terminated instances, which do not apply to RDP or SSH access. But RDP or SSH connection to an EC2 instance does generate an event in the system, such as a log entry which can be used to notify the Operation team. Since its a log, you would require a service that monitors logs like CloudTrail, VPC Flow logs, or AWS Systems Manager Session Manager.

upvoted 3 times

JayBee65 1 year, 10 months ago

I completely agree with the logic here, but I'm thinking C, since I believe you will need to "Create required metric filters" in order to detect RDP or SSH access, and this is not specified in the question, see https://docs.aws.amazon.com/systems-manager/latest/userguide/OpsCenter-create-OpsItems-from-CloudWatch-Alarms.html

upvoted 2 times

owlminus 1 year, 10 months ago

Selected Answer: C

It's C fam. RDP or SSH connections won't change the state of the EC2 instance, so D doesn't make sense.

upvoted 5 times

forzadejan 1 year, 10 months ago

D. Configure an Amazon EventBridge rule to listen for events of type EC2 Instance State-change Notification. Configure an Amazon Simple Notification Service (Amazon SNS) topic as a target. Subscribe the operations team to the topic.

EC2 instances sends events to the EventBridge when state change occurs, such as when a new RDP or SSH connection is established, you can use EventBridge to configure a rule that listens for these events and trigger an action, like sending an email or SMS, when the connection is detected. The operations team can be notified by subscribing to the Amazon Simple Notification Service (Amazon SNS) topic, which can be configured as the target of the EventBridge rule.

upvoted 3 times

alanp 1 year, 10 months ago

Are state changes pending:
running
stopping
stopped
shutting-down
terminated

https://aws.amazon.com/blogs/security/how-to-monitor-and-visualize-failed-ssh-access-attempts-to-amazon-ec2-linux-instances/

instances,
upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #233 *Topic 1*

A solutions architect has created a new AWS account and must secure AWS account root user access.

Which combination of actions will accomplish this? (Choose two.)

A. Ensure the root user uses a strong password. Most Voted

B. Enable multi-factor authentication to the root user. Most Voted

C. Store root user access keys in an encrypted Amazon S3 bucket.

D. Add the root user to a group containing administrative permissions.

E. Apply the required permissions to the root user with an inline policy document.

Correct Answer: *AB*

Community vote distribution

AB (86%) Other (14%)

Comments

cookieMr Highly Voted 1 year, 5 months ago

Selected Answer: AB

A. Setting a strong password for the root user is an essential security measure to prevent unauthorized access.

B. Enabling MFA adds an extra layer of security by requiring an additional authentication factor, such as a code from a mobile app or a hardware token, in addition to the password.

C. Root user access keys should be avoided whenever possible, and it is best to use IAM users with restricted permissions instead.

D. The root user already has unrestricted access to all resources and services in the account, so granting additional administrative permissions could increase the risk of unauthorized actions.

E. Instead, it is recommended to create IAM users with appropriate permissions and use those users for day-to-day operations, while keeping the root user secured and only using it for necessary administrative tasks.

upvoted 9 times

Ruffyit Most Recent 1 year ago

Ensure the root user uses a strong password. Enable multi-factor authentication to the root user.

upvoted 2 times

TariqKipkemei 1 year, 2 months ago

Selected Answer: AB

Ensure the root user uses a strong password. Enable multi-factor authentication to the root user.

upvoted 2 times

DiscussionMonke 1 year, 5 months ago

Selected Answer: AB

Options A & B are the CORRECT answers.

upvoted 2 times

Bmarodi 1 year, 6 months ago

Selected Answer: AB

Options A & B are the right answers.

upvoted 2 times

luisgu 1 year, 7 months ago

Selected Answer: AB

See https://docs.aws.amazon.com/SetUp/latest/UserGuide/best-practices-root-user.html

upvoted 2 times

Kunj7 1 year, 8 months ago

Selected Answer: AB

A and B are the correct answers:

Option A: A strong password is always required for any AWS account you create, and should not be shared or stored anywhere as there is always a risk.

Option B: This is following AWS best practice, by enabling MFA on your root user which provides another layer of security on the account and unauthorised access will be denied if the user does not have the correct password and MFA.

upvoted 2 times

WherecanIstart 1 year, 8 months ago

Selected Answer: AB

AB are the right answers.

upvoted 2 times

fkie4 1 year, 9 months ago

This is probably the hardest question in AWS history

upvoted 3 times

ProfXsamson 1 year, 10 months ago

Selected Answer: AB

AB is the only feasible answer here.

upvoted 4 times

bullrem 1 year, 10 months ago

Selected Answer: BE

B. Enabling multi-factor authentication for the root user provides an additional layer of security to ensure that only authorized individuals are able to access the root user account.
E. Applying the required permissions to the root user with an inline policy document ensures that the root user only has the necessary permissions to perform the necessary tasks, and not any unnecessary permissions that could potentially be misused.

upvoted 2 times

pentium75 11 months, 2 weeks ago

E is wrong because you can't attach permissions or policies to the root user.
A is right because MFA alone won't help too much if the password is "123".

upvoted 3 times

bullrem 1 year, 10 months ago

The other options are not sufficient to secure the root user access because:
A. A strong password alone is not enough to protect against potential security threats such as phishing or brute force attacks.
C. Storing the root user access keys in an encrypted S3 bucket does not address the root user's authentication process.
D. Adding the root user to a group with administrative permissions does not address the root user's authentication process and does not provide an additional layer of security.
upvoted 1 times

[Removed] 1 year, 8 months ago

Strong passwords + multi factor is the counter to brute force...
upvoted 1 times

bullrem 1 year, 10 months ago

https://docs.aws.amazon.com/IAM/latest/UserGuide/access_policies.html
upvoted 1 times

Pindol 1 year, 10 months ago

Selected Answer: AB

AB obviusly
upvoted 2 times

david76x 1 year, 10 months ago

Selected Answer: AB

Root user already has admin, so D is not correct
upvoted 1 times

Aninina 1 year, 10 months ago

Selected Answer: AB

AB are correct
upvoted 2 times

wmp7039 1 year, 10 months ago

Selected Answer: AB

D is incorrect as root user already has full admin access.
upvoted 2 times

swolfgang 1 year, 10 months ago

Selected Answer: AB

D. Add the root user to a group containing administrative permissions. >>its not about security,actually its unsecure so >> a&B
upvoted 2 times

raf123123 1 year, 10 months ago

Selected Answer: BD

BD is correct
upvoted 2 times

awsgeek75 11 months, 1 week ago

root user is literally the super admin account. What more permissions could you possible give to the root user by adding it to admin group?
upvoted 1 times

pentium75 11 months, 2 weeks ago

D is wrong because the root user is outside of IAM, thus you can't put him into a group. Also he does not need "administrative permissions" as he has those anyway.
upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #234 *Topic 1*

A company is building a new web-based customer relationship management application. The application will use several Amazon EC2 instances that are backed by Amazon Elastic Block Store (Amazon EBS) volumes behind an Application Load Balancer (ALB). The application will also use an Amazon Aurora database. All data for the application must be encrypted at rest and in transit.

Which solution will meet these requirements?

A. Use AWS Key Management Service (AWS KMS) certificates on the ALB to encrypt data in transit. Use AWS Certificate Manager (ACM) to encrypt the EBS volumes and Aurora database storage at rest.

B. Use the AWS root account to log in to the AWS Management Console. Upload the company's encryption certificates. While in the root account, select the option to turn on encryption for all data at rest and in transit for the account.

C. Use AWS Key Management Service (AWS KMS) to encrypt the EBS volumes and Aurora database storage at rest. Attach an AWS Certificate Manager (ACM) certificate to the ALB to encrypt data in transit. **Most Voted**

D. Use BitLocker to encrypt all data at rest. Import the company's TLS certificate keys to AWS Key Management Service (AWS KMS) Attach the KMS keys to the ALB to encrypt data in transit.

Correct Answer: *C*

Community vote distribution

C (100%)

Comments

cookieMr Highly Voted 1 year, 5 months ago

Selected Answer: C

AWS KMS can be used to encrypt the EBS and Aurora database storage at rest.
ACM can be used to obtain an SSL/TLS certificate and attach it to the ALB. This encrypts the data in transit between the clients and the ALB.

A is incorrect because it suggests using ACM to encrypt the EBS, which is not the correct service for encrypting EBS.

B is incorrect because relying on the AWS root account and selecting an option in the AWS Management Console to enable encryption for all data at rest and in transit is not a valid approach.

D is incorrect because BitLocker is not a suitable solution for encrypting data in AWS services. It is primarily used for encrypting data on Windows-based operating systems. Additionally, importing TLS certificate keys to AWS KMS and attaching them to the ALB is not the recommended approach for encrypting data in transit.

upvoted 11 times

Awsbeginner87 Most Recent 8 months ago

Got this question in exam today

upvoted 4 times

Ruffyit 1 year ago

To encrypt data at rest, AWS Key Management Service (AWS KMS) can be used to encrypt EBS volumes and Aurora database storage.

To encrypt data in transit, an AWS Certificate Manager (ACM) certificate can be attached to the Application Load Balancer (ALB) to enable HTTPS and TLS encryption.

upvoted 1 times

TariqKipkemei 1 year, 2 months ago

Selected Answer: C

Use AWS Key Management Service (AWS KMS) to encrypt the EBS volumes and Aurora database storage at rest. Attach an AWS Certificate Manager (ACM) certificate to the ALB to encrypt data in transit

upvoted 2 times

Guru4Cloud 1 year, 2 months ago

Selected Answer: C

C is the best answer.

To encrypt data at rest, AWS Key Management Service (AWS KMS) can be used to encrypt EBS volumes and Aurora database storage.

To encrypt data in transit, an AWS Certificate Manager (ACM) certificate can be attached to the Application Load Balancer (ALB) to enable HTTPS and TLS encryption.

upvoted 3 times

MAMADOUG 1 year, 5 months ago

Selected Answer: C

Option C it's correct

upvoted 2 times

Bmarodi 1 year, 6 months ago

Selected Answer: C

Option C fulfills the requirements.

upvoted 2 times

techhb 1 year, 10 months ago

Selected Answer: C

C is correct ,A REVERSES the work ofeach service.

upvoted 4 times

Aninina 1 year, 10 months ago

Selected Answer: C

C is correct!

upvoted 4 times

mhmt4438 1 year, 10 months ago

Selected Answer: C

c is correct answer

upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #235 *Topic 1*

A company is moving its on-premises Oracle database to Amazon Aurora PostgreSQL. The database has several applications that write to the same tables. The applications need to be migrated one by one with a month in between each migration. Management has expressed concerns that the database has a high number of reads and writes. The data must be kept in sync across both databases throughout the migration.

What should a solutions architect recommend?

A. Use AWS DataSync for the initial migration. Use AWS Database Migration Service (AWS DMS) to create a change data capture (CDC) replication task and a table mapping to select all tables.

B. Use AWS DataSync for the initial migration. Use AWS Database Migration Service (AWS DMS) to create a full load plus change data capture (CDC) replication task and a table mapping to select all tables.

C. Use the AWS Schema Conversion Tool with AWS Database Migration Service (AWS DMS) using a memory optimized replication instance. Create a full load plus change data capture (CDC) replication task and a table mapping to select all tables. Most Voted

D. Use the AWS Schema Conversion Tool with AWS Database Migration Service (AWS DMS) using a compute optimized replication instance. Create a full load plus change data capture (CDC) replication task and a table mapping to select the largest tables.

Correct Answer: *C*

Community vote distribution

C (92%) A (8%)

Comments

aakashkumar1999 Highly Voted 1 year, 10 months ago

Selected Answer: C

C : because we need SCT to convert from Oracle to PostgreSQL, and we need memory optimized machine for databases not compute optimized.

upvoted 16 times

hissein 1 year, 3 months ago

why it is memory optimized and not compute optimized machine ?

upvoted 8 times

Guru4Cloud 1 year, 2 months ago

A memory-optimized replication instance is recommended because the database has a high number of reads and writes. Memory-optimized instances are designed to deliver fast performance for workloads that process large data sets in memory.

upvoted 22 times

hissein 1 year, 2 months ago

thank you

upvoted 2 times

pentium75 11 months, 2 weeks ago

Maybe it doesn't matter, but in D we create a table mapping only for "the largest tables" while obviously we need "all tables" as in C.

upvoted 8 times

TechStuff Most Recent 9 months, 3 weeks ago

B. Use AWS DataSync for the initial migration. Use AWS Database Migration Service (AWS DMS) to create a full load plus change data capture (CDC) replication task and a table mapping to select all tables.

This approach leverages AWS DataSync for the initial data migration, which is optimized for high-speed transfer of large amounts of data. Then, AWS DMS is used to create a replication task with full load plus change data capture (CDC), ensuring ongoing synchronization between the on-premises Oracle database and Amazon Aurora PostgreSQL. By selecting all tables, the migration process ensures that all applications can continue to read from and write to the database without interruption during the migration period.

upvoted 2 times

MrPCarrot 9 months, 4 weeks ago

Answer is C - AWS Schema Conversion Tool (AWS SCT) supports heterogeneous database migrations by automatically converting the source database schema and a majority of the custom code to a format compatible with the target database.

upvoted 1 times

awsgeek75 10 months, 3 weeks ago

Selected Answer: C

Has to be SCT + DMS for all the tables so C is the choice. Why do you need SCT? Read this:
https://docs.aws.amazon.com/prescriptive-guidance/latest/patterns/migrate-data-from-an-on-premises-oracle-database-to-aurora-postgresql.html

upvoted 2 times

awsgeek75 10 months, 3 weeks ago

AB-> DataSync has nothing to do with DB Migration (https://aws.amazon.com/datasync/)
D: Only migrates the largest table

I think these questions are more for seeing how much of AWS product catalogue can you remember efficiently with associated features. Afterall AWS SA is also a salesperson for the right product! I now look at product cheat-sheets to look them up at work.

upvoted 2 times

pentium75 11 months, 2 weeks ago

Selected Answer: C

Oracle -> PostgreSQL, we need SCT, thus A and B are out.
D maps only "the largest tables" but we need all tables

upvoted 3 times

ansagr 12 months ago

Selected Answer: C

Another reason to rule out D is because it states “a table mapping to select the largest tables”, whereas selecting all tables (as stated in option C) in the table mapping is necessary to ensure a comprehensive migration.

upvoted 4 times

Ruffyit 1 year ago

because we need SCT to convert from Oracle to PostgreSQL, and we need memory optimized machine for databases not compute optimized.
A memory-optimized replication instance is recommended because the database has a high number of reads and writes. Memory-optimized instances are designed to deliver fast performance for workloads that process large data sets in memory.

upvoted 2 times

Po_chih 1 year, 2 months ago

Selected Answer: C

because we need SCT to convert from Oracle to PostgreSQL, and we need memory optimized machine for databases not compute optimized.
https://repost.aws/zh-Hant/knowledge-center/dms-optimize-aws-sct-performance

upvoted 2 times

TariqKipkemei 1 year, 2 months ago

Selected Answer: C

Oracle database to Amazon Aurora PostgreSQL = AWS Schema Conversion Tool
High number of reads and writes = memory optimized replication instance

upvoted 3 times

Guru4Cloud 1 year, 2 months ago

Selected Answer: C

A memory-optimized replication instance is recommended because the database has a high number of reads and writes. Memory-optimized instances are designed to deliver fast performance for workloads that process large data sets in memory.

upvoted 2 times

_d1rk_ 1 year, 3 months ago

Selected Answer: C

DataSync is for file-level synch, so A and B can be excluded. C is better than D because memory-optimized instances are recommended to handle the high number of reads and writes

upvoted 2 times

ukivanlamlpi 1 year, 3 months ago

Selected Answer: A

why not a? only capture the change is sufficient

upvoted 2 times

pentium75 11 months, 2 weeks ago

Oracle -> PostgreSQL requires SCT

upvoted 1 times

Mmmmmmkkkk 1 year, 5 months ago

Bbbbbb

upvoted 1 times

cookieMr 1 year, 5 months ago

Selected Answer: C

The AWS SCT is used to convert the schema and code of the Oracle database to be compatible with Aurora PostgreSQL. AWS DMS is utilized to migrate the data from the Oracle database to Aurora PostgreSQL. Using a memory-optimized replication instance is recommended to handle the high number of reads and writes during the migration process.
By creating a full load plus CDC replication task, the initial data migration is performed, and ongoing changes in the Oracle database are continuously captured and applied to the Aurora PostgreSQL database. Selecting all tables for table mapping ensures that all the applications writing to the same tables are migrated.

Option A & B are incorrect because using AWS DataSync alone is not sufficient for database migration and data synchronization.

Option D is incorrect because using a compute optimized replication instance is not the most suitable choice for handling the high number of reads and writes.

upvoted 3 times

omoakin 1 year, 6 months ago

BBBBBBBBBBBBB

upvoted 2 times

SimiTik 1 year, 7 months ago

B chatgpt

upvoted 2 times

KZM 1 year, 9 months ago

DMS+SCT for Oracle to Aurora PostgreSQL migration
https://docs.aws.amazon.com/prescriptive-guidance/latest/patterns/migrate-an-oracle-database-to-aurora-postgresql-using-aws-dms-and-aws-sct.html

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #236 *Topic 1*

A company has a three-tier application for image sharing. The application uses an Amazon EC2 instance for the front-end layer, another EC2 instance for the application layer, and a third EC2 instance for a MySQL database. A solutions architect must design a scalable and highly available solution that requires the least amount of change to the application.

Which solution meets these requirements?

A. Use Amazon S3 to host the front-end layer. Use AWS Lambda functions for the application layer. Move the database to an Amazon DynamoDB table. Use Amazon S3 to store and serve users' images.

B. Use load-balanced Multi-AZ AWS Elastic Beanstalk environments for the front-end layer and the application layer. Move the database to an Amazon RDS DB instance with multiple read replicas to serve users' images.

C. Use Amazon S3 to host the front-end layer. Use a fleet of EC2 instances in an Auto Scaling group for the application layer. Move the database to a memory optimized instance type to store and serve users' images.

D. Use load-balanced Multi-AZ AWS Elastic Beanstalk environments for the front-end layer and the application layer. Move the database to an Amazon RDS Multi-AZ DB instance. Use Amazon S3 to store and serve users' images. **Most Voted**

Correct Answer: *D*

Community vote distribution

D (75%) B (23%) A

Comments

PDR Highly Voted 1 year, 10 months ago

Selected Answer: B

B and D very similar with D being the 'best' solution but it is not the one that requires the least amount of development changes as the application would need to be changed to store images in S3 instead of DB

upvoted 15 times

pentium75 11 months, 2 weeks ago

B is wrong because single "RDS DB instance" is not HA.

No one says that the images are currently stored in S3. Also the requirement is "least amount of change [not "no change"] to the application".

upvoted 8 times

Aninina Highly Voted 1 year, 10 months ago

Selected Answer: D

for "Highly available": Multi-AZ &
for "least amount of changes to the application": Elastic Beanstalk automatically handles the deployment, from capacity provisioning, load balancing, auto-scaling to application health monitoring

upvoted 12 times

SirDNS Most Recent 2 months, 2 weeks ago

Selected Answer: D

When we are talking about sharing static content and options have S3 why would I select any other option.

upvoted 1 times

satyaammm 3 months, 2 weeks ago

Selected Answer: D

D is the most suitable option here.

upvoted 1 times

awsgeek75 11 months, 1 week ago

Selected Answer: D

A: Requires changing EC2 application to Lambda. Seems like a big change
B: RDS DB is not best option for serve images and also single instance isn't HA
C: Memory optimised instance is not HA
D: Multi-AZ EBS is lift and shift for EC2 front-end and app later. RDS Multi AZ is HA. S3 for static images is best performance/scalability/availability.

upvoted 4 times

ansagr 12 months ago

Selected Answer: D

Using Amazon RDS for serving images might not be the optimal solution, as RDS is more suitable for storing structured data in a relational database rather than BLOBs like images. Storing and serving images can be more efficiently handled by object storage services like Amazon S3.

upvoted 2 times

Ruffyit 1 year ago

Use load-balanced Multi-AZ AWS Elastic Beanstalk environments for the front-end layer and the application layer. Move the database to an Amazon RDS Multi-AZ DB instance. Use Amazon S3 to store and serve users' images

upvoted 2 times

rlamberti 1 year, 1 month ago

Option B - DB is not a good option to store images. Read replicas won't improve HA for write, only scalates reading IO. Therefore no true HA achieved.
D is the goal for me.

upvoted 2 times

TariqKipkemei 1 year, 2 months ago

Selected Answer: D

Use load-balanced Multi-AZ AWS Elastic Beanstalk environments for the front-end layer and the application layer. Move the database to an Amazon RDS Multi-AZ DB instance. Use Amazon S3 to store and serve users' images

upvoted 3 times

Guru4Cloud 1 year, 2 months ago

Selected Answer: D

Use Elastic Beanstalk load-balanced environments for the web and app tiers. This provides auto scaling and high availability with minimal effort.
Move the database to RDS Multi-AZ. This handles scaling reads and storage, and provides HA with automated failover.
Use S3 for serving user images. S3 is highly scalable and durable storage.
The application code remains unchanged using this approach.

upvoted 4 times

upvoted 4 times

Mia2009687 1 year, 5 months ago

Selected Answer: A

AWS Elastic Beanstalk makes it even easier for developers to quickly deploy and manage applications in the AWS Cloud. Developers simply upload their application, and Elastic Beanstalk automatically handles the deployment details of capacity provisioning, load balancing, auto-scaling, and application health monitoring.

I don't quite understand why people choose D.

upvoted 1 times

cookieMr 1 year, 5 months ago

Selected Answer: D

By using load-balanced Multi-AZ AWS EBS, you achieve scalability and high availability for both layers without requiring significant changes to the application. Moving the DB to an RDS Multi-AZ DB ensures high availability and automatic failover. Storing and serving users' images through S3 provides a scalable and highly available solution.

A is incorrect because using S3 for the front-end layer and Lambda for the application layer would require significant changes to the application architecture. Moving the DB to DynamoDB would require rewriting the DB-related code.

B is incorrect because using load-balanced Multi-AZ AWS EBS environments and an RDS DB with read replicas for serving images would be a more suitable solution. RDS with read replicas can handle the image-serving workload more efficiently than using S3 for this purpose.

C is incorrect because using S3 for the front-end layer and an ASG of EC2 for the application layer would require modifying the application architecture. Storing and serving images from a memory-optimized EC2 type may not be the most efficient and scalable approach compared to using S3.

upvoted 5 times

markw92 1 year, 5 months ago

"least amount of change to the application." - A has lots of changes, completely revamping the application and lots of new pieces. D is closest with only addition of s3 to store images which is right move. You do not want images to store in any database anyway.

upvoted 4 times

aaroncelestin 1 year, 3 months ago

Thats what I was thinking, but the question doesn't mention anything about storing users' images anywhere. Are we supposed to just assume that they wanted to store the images in a DB even though that is a bad idea?

upvoted 2 times

Bmarodi 1 year, 6 months ago

Selected Answer: D

Option D meets the requirements.

upvoted 2 times

Grace83 1 year, 8 months ago

D is correct

upvoted 3 times

focus_23 1 year, 10 months ago

Selected Answer: D

RDS multi AZ.

upvoted 3 times

wmp7039 1 year, 10 months ago

Selected Answer: D

D is correct as application changes needs to me minimal

upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #237 *Topic 1*

An application running on an Amazon EC2 instance in VPC-A needs to access files in another EC2 instance in VPC-B. Both VPCs are in separate AWS accounts. The network administrator needs to design a solution to configure secure access to EC2 instance in VPC-B from VPC-A. The connectivity should not have a single point of failure or bandwidth concerns.

Which solution will meet these requirements?

A. Set up a VPC peering connection between VPC-A and VPC-B. **Most Voted**

B. Set up VPC gateway endpoints for the EC2 instance running in VPC-B.

C. Attach a virtual private gateway to VPC-B and set up routing from VPC-A.

D. Create a private virtual interface (VIF) for the EC2 instance running in VPC-B and add appropriate routes from VPC-A.

Correct Answer: *A*

Community vote distribution

A (97%) B

Comments

cookieMr Highly Voted 1 year, 11 months ago

Selected Answer: A

A VPC peering connection allows secure communication between instances in different VPCs using private IP addresses without the need for internet gateways, VPN connections, or NAT devices. By setting it up, the application running in VPC-A can directly access the EC2 in VPC-B without going through the public internet or any single point of failure.

B is incorrect because VPC gateway endpoints are used for accessing S3 or DynamoDB from a VPC without going over the internet. They are not designed for establishing connectivity between EC2 instances in different VPCs.

C is incorrect because it would require configuring a VPN connection between the VPCs. This would introduce additional complexity and potential single points of failure.

D is incorrect because creating a private VIF and adding routes would be applicable for establishing a direct connection between on-premises infrastructure and VPC-B using Direct Connect, but it is not suitable for the scenario of communication between EC2 instances in separate VPCs within different AWS accounts.

upvoted 19 times

LuckyAro Highly Voted 2 years, 4 months ago

Selected Answer: A

AWS uses the existing infrastructure of a VPC to create a VPC peering connection; it is neither a gateway nor a VPN connection, and does not rely on a separate piece of physical hardware. There is no single point of failure for communication or a bandwidth bottleneck.

https://docs.aws.amazon.com/vpc/latest/peering/what-is-vpc-peering.html
upvoted 19 times

satyaammm Most Recent 3 months, 2 weeks ago

Selected Answer: A

VPC peering is the most suitable here.
upvoted 1 times

Faridtnx 1 year, 2 months ago

Selected Answer: A

You can create a VPC peering connection between your own VPCs, or with a VPC in another AWS account. Peering within the same AZ is free of charge.
upvoted 3 times

lostmagnet001 1 year, 4 months ago

Selected Answer: A

I get a little confused about B and A but, because, with a VPC endpoint in B it will work too access from A.
upvoted 2 times

GPFT 8 months, 4 weeks ago

vpc gw endpoint just dynamo and s3
upvoted 2 times

pentium75 1 year, 5 months ago

Selected Answer: A

B is wrong because "VPC gateway endpoint" is for S3 or DynamoDB, not EC2
C is overkill, would require a second gateway in VPC-A, not be HA and have limited bandwidth
D is wrong because VIF is for Direct Connect, has nothing to do with VPC-to-VPC communication
upvoted 5 times

Ruffyit 1 year, 6 months ago

AWS uses the existing infrastructure of a VPC to create a VPC peering connection; it is neither a gateway nor a VPN connection, and does not rely on a separate piece of physical hardware. There is no single point of failure for communication or a bandwidth bottleneck.

https://docs.aws.amazon.com/vpc/latest/peering/what-is-vpc-peering.html
upvoted 2 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: A

A. Set up a VPC peering connection between VPC-A and VPC-B
upvoted 2 times

MNotABot 1 year, 10 months ago

https://www.bing.com/search?pglt=41&q=can+we+do+VPC+peering+across+AWS+accounts&cvid=48a8ceecc85a429c9ddd698b01055890&aqs=edge..69i57j0l8j69i11004.10897j0j1&FORM=ANNAB1&PC=LCTS
upvoted 1 times

Anmol_1010 1 year, 11 months ago

D, VPC PEERINGVIS IN SAME ACCOUNT
upvoted 1 times

im6h 1 year, 11 months ago

No, VPC Peering can use across account.

https://docs.aws.amazon.com/vpc/latest/peering/what-is-vpc-peering.html
upvoted 4 times

omoakin 2 years ago

DDDDDDDDDDDDDD
upvoted 2 times

omoakin 2 years ago

This is the only viable solution
Create a private virtual interface (VIF) for the EC2 instance running in VPC-B and add appropriate routes from VPC-A
upvoted 1 times

michellemeloc 2 years ago

Selected Answer: A

"You can create a VPC peering connection between your own VPCs, or with a VPC in another AWS account."

https://docs.aws.amazon.com/vpc/latest/peering/what-is-vpc-peering.html
upvoted 5 times

PDR 2 years, 4 months ago

Selected Answer: A

correct answer is A and as mentioned by JayBee65 below, key reason being that solution should not have a single point of failure and bandwidth restrictions

the following paragraph is taken from the AWS docs page linked below that backs this up
"AWS uses the existing infrastructure of a VPC to create a VPC peering connection; it is neither a gateway nor a VPN connection, and does not rely on a separate piece of physical hardware. There is no single point of failure for communication or a bandwidth bottleneck."

https://docs.aws.amazon.com/vpc/latest/peering/what-is-vpc-peering.html
upvoted 3 times

LuckyAro 2 years, 4 months ago

Selected Answer: B

A VPC endpoint gateway to the EC2 Instance is more specific and more secure than forming a VPC peering that exposes the whole of the VPC infrastructure just for one connection.
upvoted 2 times

pentium75 1 year, 5 months ago

B is about a Gateway endpoint, which can be used to connect to S3 or DynamoDB, NOT to another EC2 instance.
upvoted 2 times

JayBee65 2 years, 4 months ago

Your logic is correct but security is not a requirement here - the requirements are "The connectivity should not have a single point of failure or bandwidth concerns." A VPC gateway endpoint" would form a single point of failure, so B is incorrect, (and C and D are incorrect for the same reason, they create single points of failure).
upvoted 4 times

mhmt4438 2 years, 4 months ago

Selected Answer: A

Correct answer is A
upvoted 3 times

Aninina 2 years, 4 months ago

Selected Answer: A

VPC peering allows resources in different VPCs to communicate with each other as if they were within the same network. This solution would establish a direct network route between VPC-A and VPC-B, eliminating the need for a single point of failure or bandwidth concerns.

upvoted 2 times

waiyiu9981 2 years, 4 months ago

Selected Answer: A

https://www.examtopics.com/discussions/amazon/view/27763-exam-aws-certified-solutions-architect-associate-saa-c02/

upvoted 5 times

EXAMTOPICS

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #238 *Topic 1*

A company wants to experiment with individual AWS accounts for its engineer team. The company wants to be notified as soon as the Amazon EC2 instance usage for a given month exceeds a specific threshold for each account.

What should a solutions architect do to meet this requirement MOST cost-effectively?

A. Use Cost Explorer to create a daily report of costs by service. Filter the report by EC2 instances. Configure Cost Explorer to send an Amazon Simple Email Service (Amazon SES) notification when a threshold is exceeded.

B. Use Cost Explorer to create a monthly report of costs by service. Filter the report by EC2 instances. Configure Cost Explorer to send an Amazon Simple Email Service (Amazon SES) notification when a threshold is exceeded.

C. Use AWS Budgets to create a cost budget for each account. Set the period to monthly. Set the scope to EC2 instances. Set an alert threshold for the budget. Configure an Amazon Simple Notification Service (Amazon SNS) topic to receive a notification when a threshold is exceeded. **Most Voted**

D. Use AWS Cost and Usage Reports to create a report with hourly granularity. Integrate the report data with Amazon Athena. Use Amazon EventBridge to schedule an Athena query. Configure an Amazon Simple Notification Service (Amazon SNS) topic to receive a notification when a threshold is exceeded.

Correct Answer: *C*

Community vote distribution

C (95%) D (5%)

Comments

Aninina Highly Voted 1 year, 4 months ago

Selected Answer: C

AWS Budgets allows you to create budgets for your AWS accounts and set alerts when usage exceeds a certain threshold. By creating a budget for each account, specifying the period as monthly and the scope as EC2 instances, you can effectively track the EC2 usage for each account and be notified when a threshold is exceeded. This solution is the most cost-effective option as it does not require additional resources such as Amazon Athena or Amazon EventBridge.

upvoted 14 times

alexleely Highly Voted 1 year, 4 months ago

C: AWS Budgets allows you to set a budget for costs and usage for your accounts and you can set alerts when the budget threshold is exceeded in real-time which meets the requirement.

Why not B: B would be the most cost-effective if the requirements didn't ask for real-time notification. You would not incur additional costs for the daily or monthly reports and the notifications. But doesn't provide real-time alerts.

upvoted 5 times

satyaammm Most Recent 3 months, 2 weeks ago

Selected Answer: C

AWS budgets allow for creating budgets and hence are the most suitable here.

upvoted 1 times

Ruffyit 6 months, 2 weeks ago

AWS Budgets allows you to create budgets for your AWS accounts and set alerts when usage exceeds a certain threshold. By creating a budget for each account, specifying the period as monthly and the scope as EC2 instances, you can effectively track the EC2 usage for each account and be notified when a threshold is exceeded. This solution is the most cost-effective option as it does not require additional resources such as Amazon Athena or Amazon EventBridge.

upvoted 1 times

vijaykamal 8 months, 1 week ago

Selected Answer: C

Option A and Option B suggest using Cost Explorer to create reports and send notifications. While Cost Explorer is useful for analyzing costs, it does not provide the real-time alerting capability that AWS Budgets offers.

Option D suggests using AWS Cost and Usage Reports integrated with Amazon Athena and Amazon EventBridge, which can be a more complex and potentially costlier solution compared to AWS Budgets for this specific use case. It's also more suitable for fine-grained, custom analytics rather than straightforward threshold-based alerts.

upvoted 4 times

TariqKipkemei 8 months, 2 weeks ago

Selected Answer: C

AWS Budgets was designed to handle this scenario.

upvoted 3 times

Undisputed 10 months, 2 weeks ago

Selected Answer: C

Use AWS Budgets to create a cost budget for each account. Set the period to monthly. Set the scope to EC2 instances. Set an alert threshold for the budget. Configure an Amazon Simple Notification Service (Amazon SNS) topic to receive a notification when a threshold is exceeded.

upvoted 2 times

cookieMr 11 months, 2 weeks ago

Selected Answer: C

By creating a cost budget for each account, specifying the period as monthly and scoping it to EC2, you can track and monitor the costs associated with EC2 specifically. Set an alert threshold in the budget, which will trigger a notification when the specified threshold is exceeded. Configure an SNS to receive the notification, which can be subscribed to by the company to receive immediate alerts.

A and B are not the most cost-effective solutions as they involve using Cost Explorer to create reports, which may not provide real-time notifications when the threshold is exceeded. Additionally, A. suggests using a daily report, while B. suggests using a monthly report, which may not provide the desired level of granularity for immediate notifications.

D involves using Cost and Usage Reports with Athena and EventBridge. This solution provides more flexibility and data analysis capabilities, it is more complex and may incur additional costs for using Athena and generating hourly reports.

upvoted 3 times

Samuel03 1 year, 3 months ago

Selected Answer: D

I go with D. It says "as soon as", "daily" reports seems to be a bit longer time frame to wait in my opinion.

upvoted 2 times

Bofi 1 year, 3 months ago

Athena can only be use in s3, that is enough to discard D

upvoted 2 times

Samuel03 1 year, 3 months ago

Actually, I take that back. It clearly says "Cost effective."

upvoted 4 times

mp165 1 year, 4 months ago

Selected Answer: C

Agree...C

upvoted 3 times

mhmt4438 1 year, 4 months ago

Selected Answer: C

Answer is C

upvoted 2 times

venice1234 1 year, 4 months ago

Selected Answer: C

https://aws.amazon.com/getting-started/hands-on/control-your-costs-free-tier-budgets/

upvoted 2 times

Morinator 1 year, 4 months ago

Selected Answer: C

AWS budget IMO, it's done for it

upvoted 3 times